

Block Diagram

Outlines

- Basics of Block Diagram
- Terms of Block Diagram
- Applications of Block Diagram
- Advantages of Block Diagram
- Disadvantages of Block Diagram

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Basics of Block Diagram

- Block Diagram is a pictorial representation of the entire System.
- Block Diagram represents the relationship between the input and output of the entire system.
- Complicated Systems can be easily represented by connecting different blocks.
- It explains interrelationships that exist among various components.
- Block Diagram represents the transfer function of a subsystem present in the Control System.
- Lines connecting the blocks are called branches.
- Arrows associated with the branches indicate the direction of the signal flow.

Terms of Block Diagram

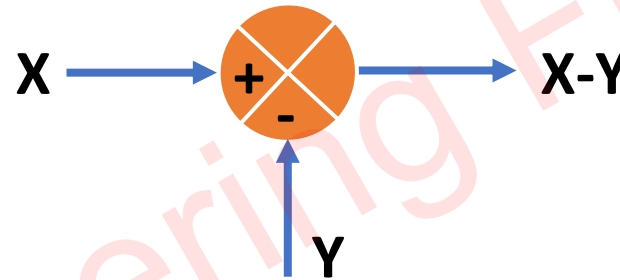
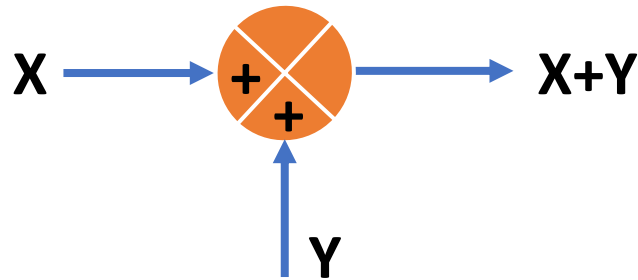
□ Output

- $\text{Output} = \text{Gain} \times \text{Input}$



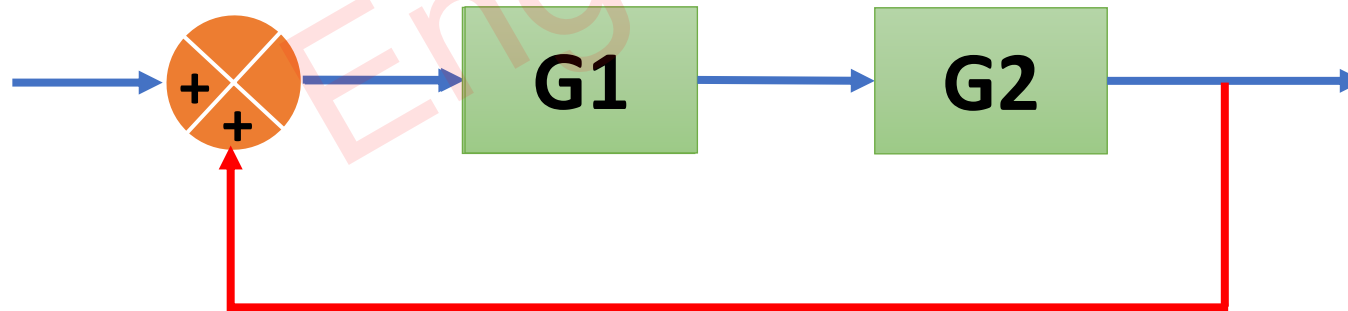
□ Summing Point

- More than one signal can be added or subtracted at the Summing point



□ Take off Point

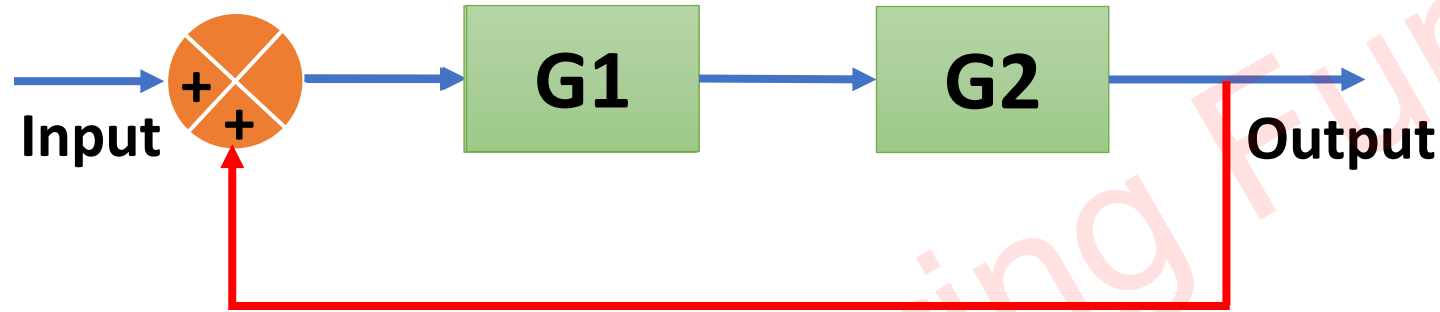
- The point from which a signal is taken for any use is called the take-off point.



Terms of Block Diagram

□ Forward Path and Reverse Path {Feedback Path}

- The Path from Input to Output is Forward Path.
- The Path from Output to Input is Feedback Path.



Applications of Block Diagram

- Hardware Design
- Electric System Design
- Software Design
- Process Flow Diagram
- In PLC SCADA System

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Advantages of Block Diagram

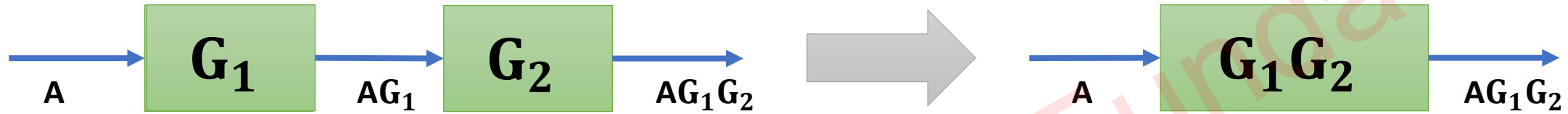
- The functional operation of the system can be observed from the block Diagram.
- Block Diagram gives the information about the performance of the system.
- A block diagram is used to analyze and design the control system.
- Block Diagram bisects the complicated system into subsystems with blocks.

Disadvantages of Block Diagram

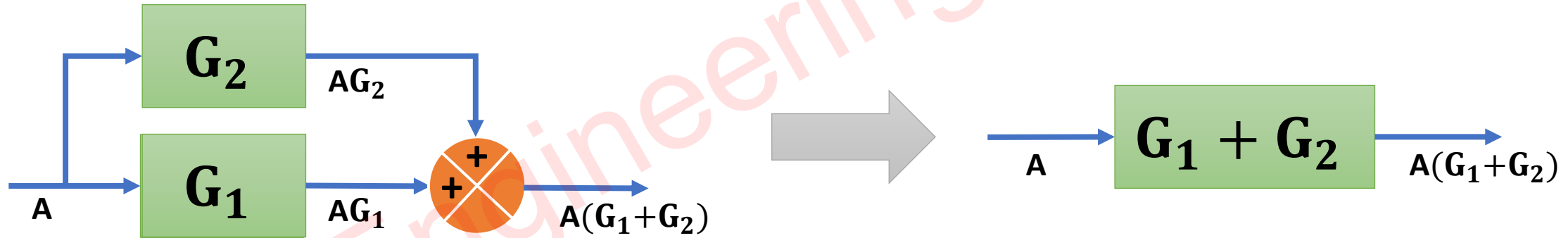
- The Block Diagram for any system is not unique.
- Source of Energy in the system is not shown in the diagram.
- In the process of block diagram reduction, some important functions may be hidden.
- It does not give information about the physical construction of the system.

Block Diagram Reduction Rules

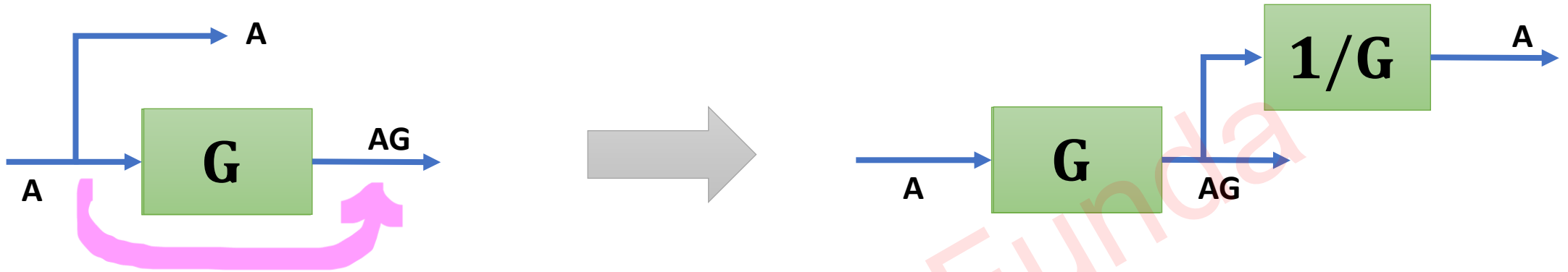
□ Blocks in Series:



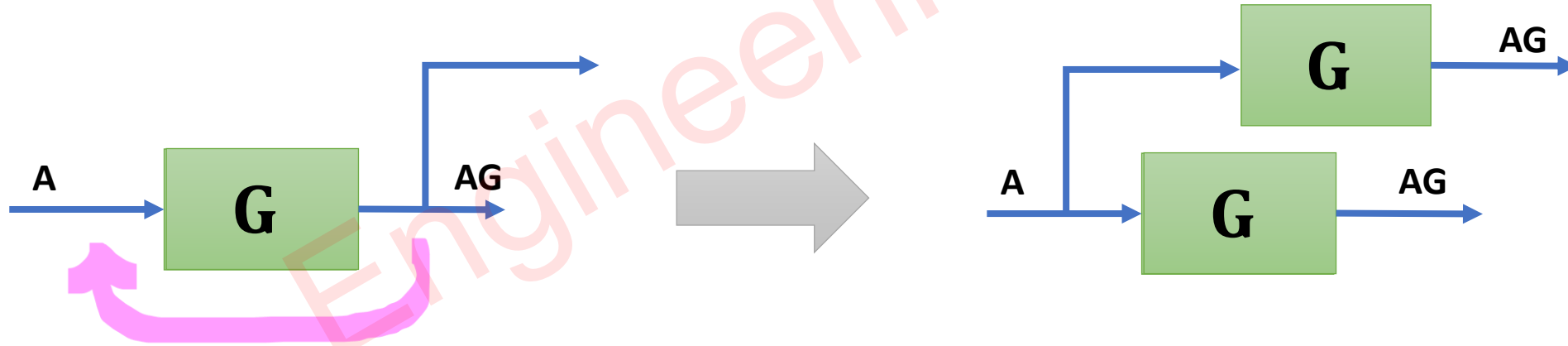
□ Blocks in Parallel:



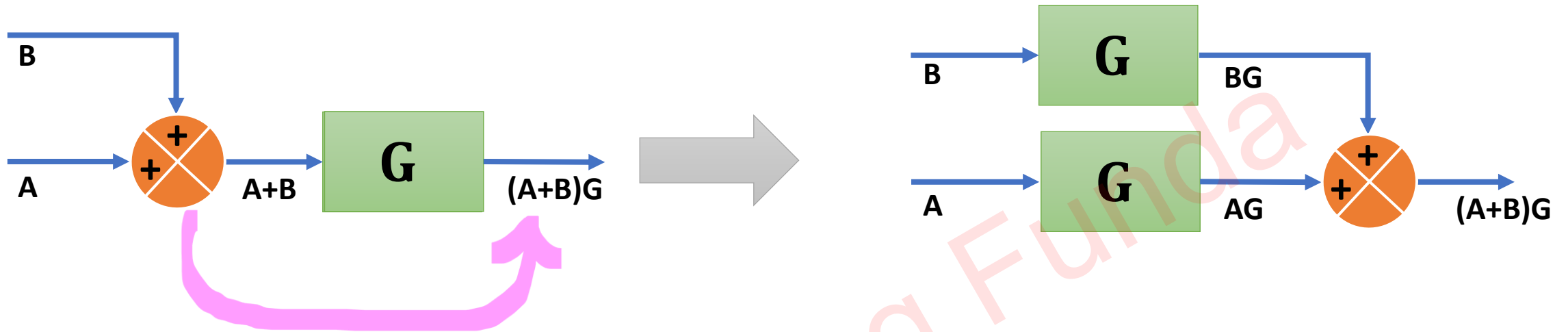
❑ Jumping Branch Point Ahead of Block:



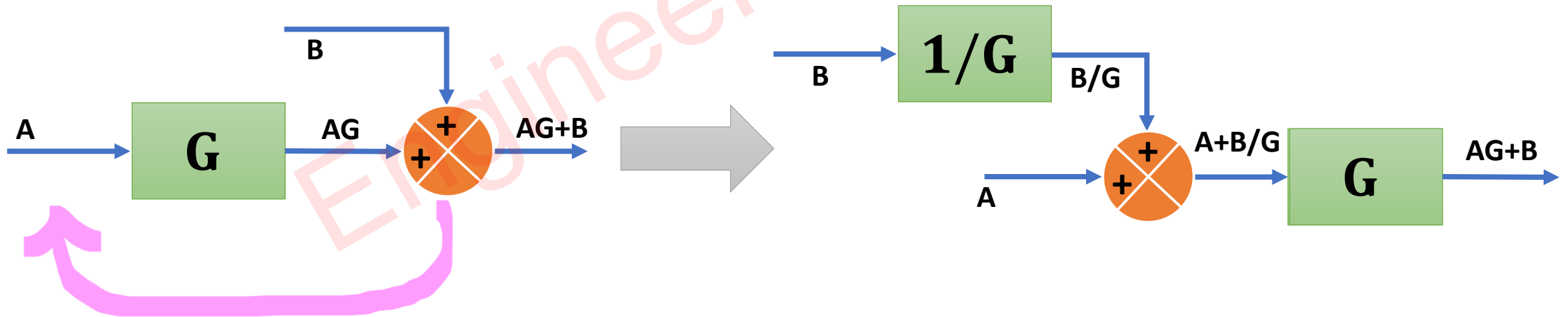
❑ Jumping Branch Point Behind of Block:



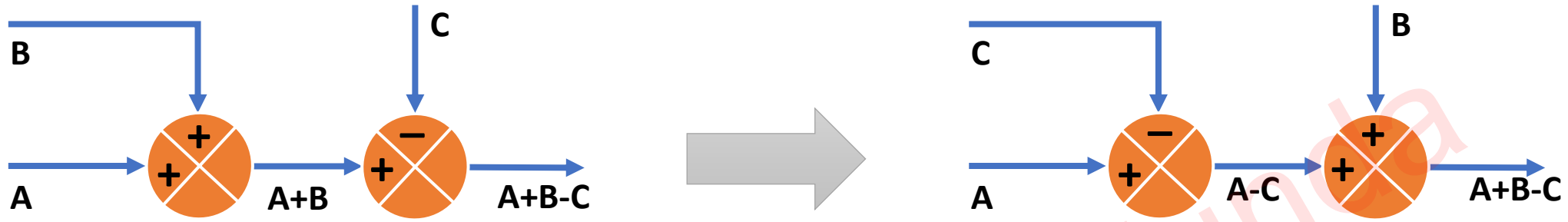
□ Jumping Summing Point Ahead of Block:



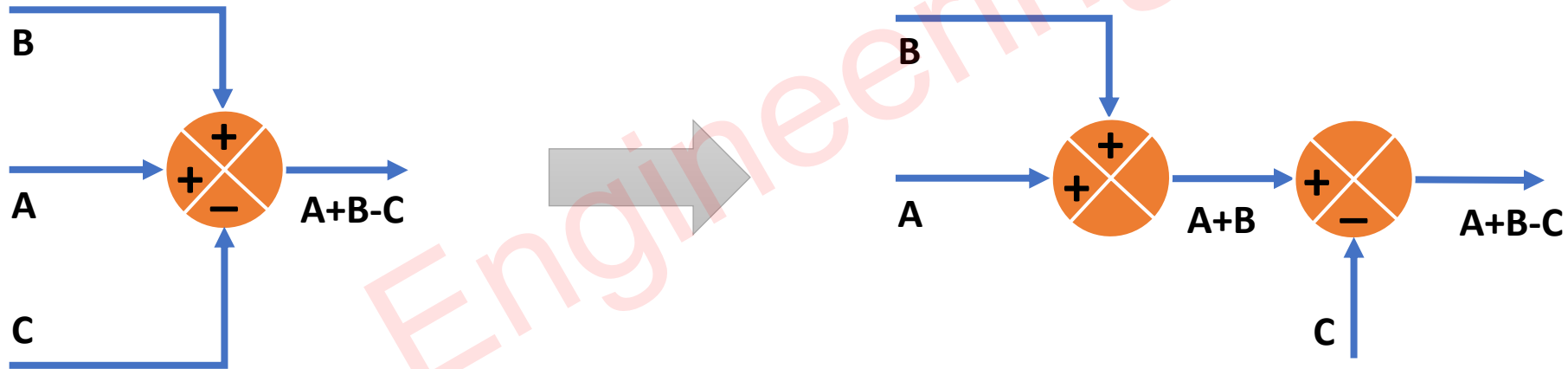
□ Jumping Summing Point Behind of Block:



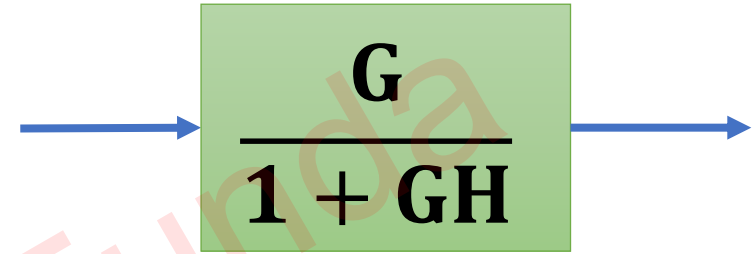
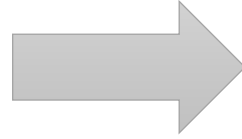
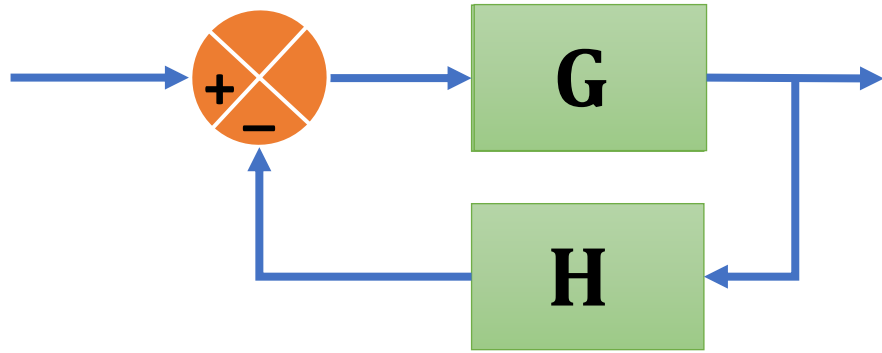
❑ Interchanging the Summing Point:



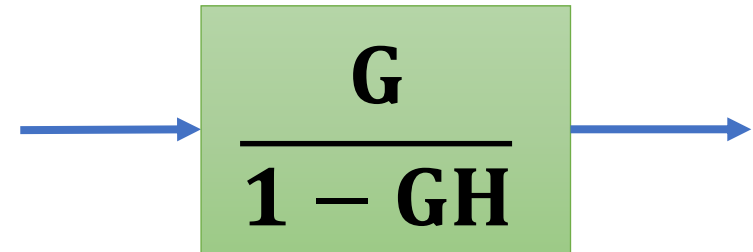
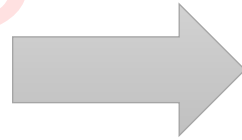
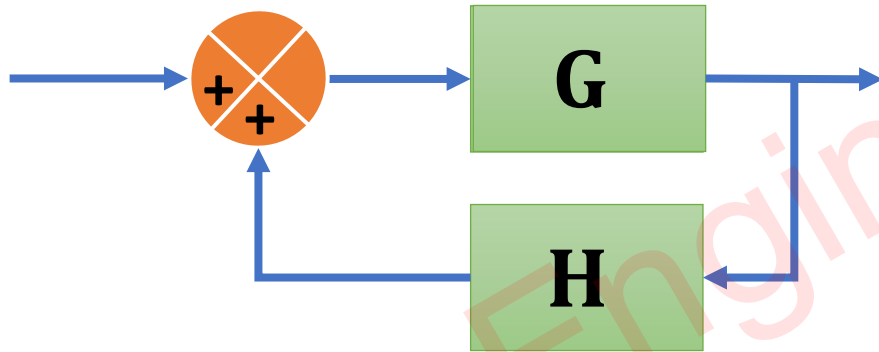
❑ Splitting or Combining the Summing Point:



❑ Negative Feedback:



❑ Positive Feedback:

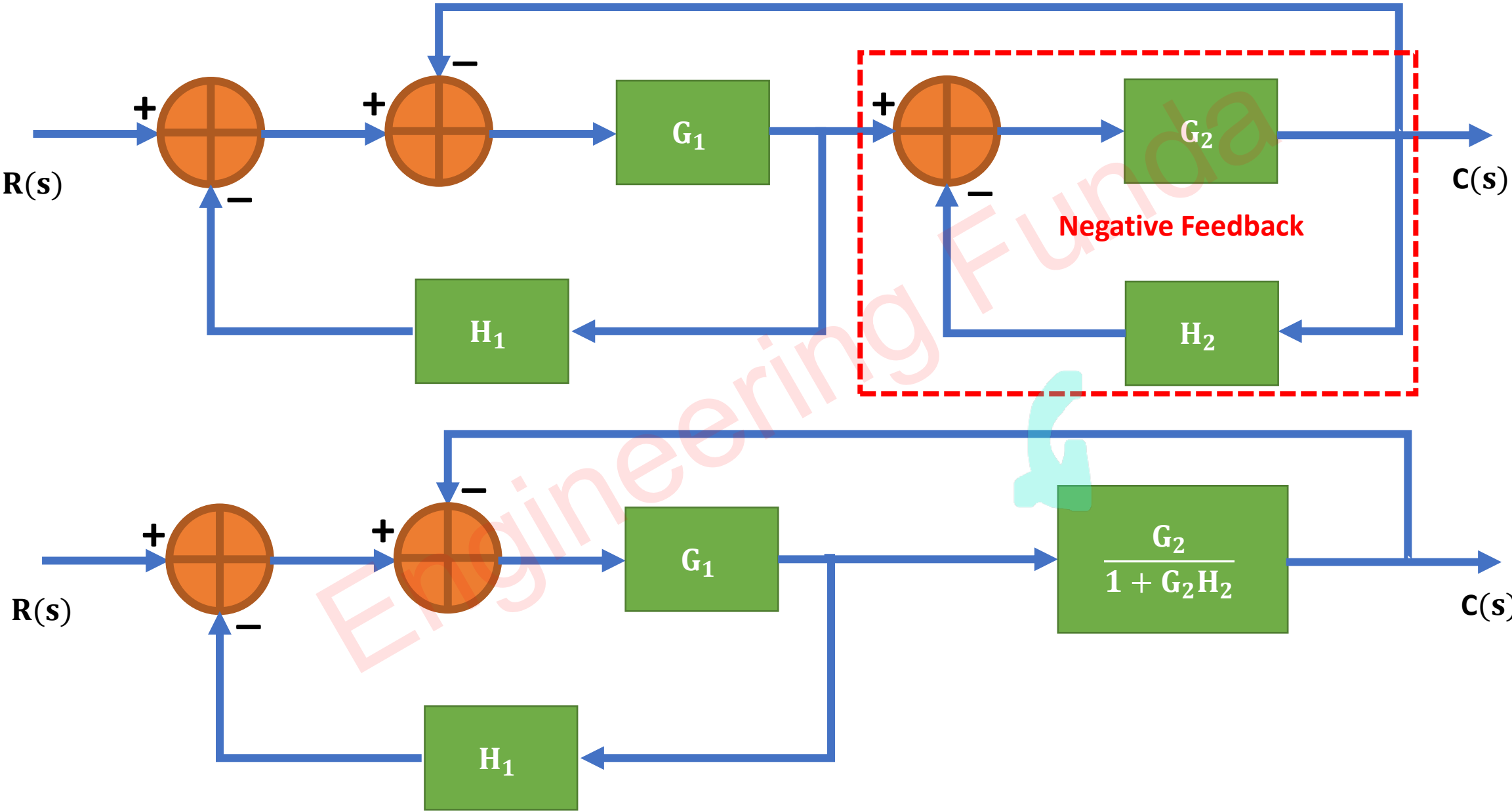


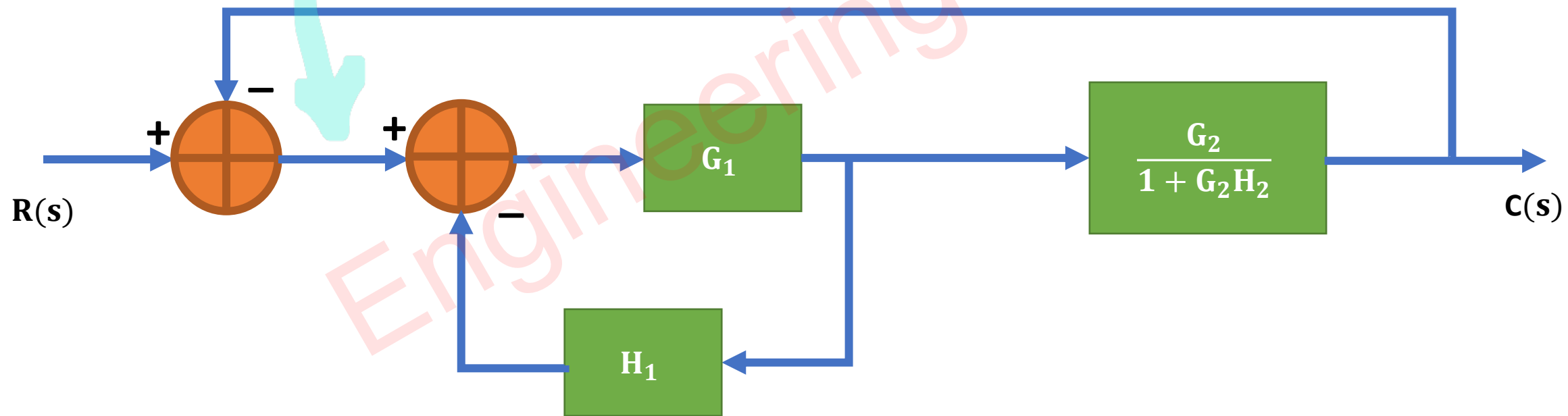
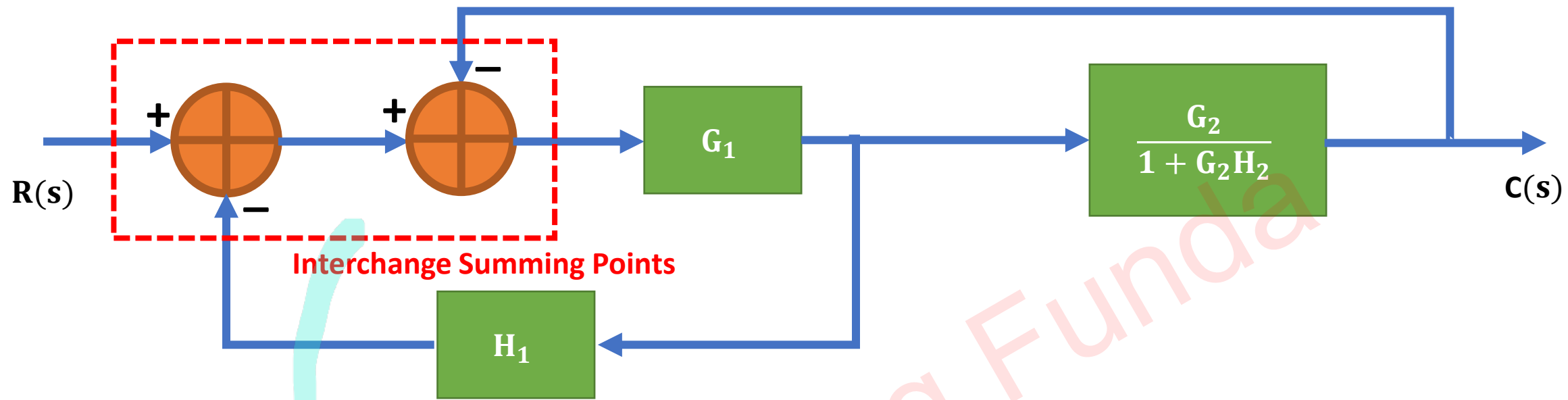
Block Diagram Reduction Examples

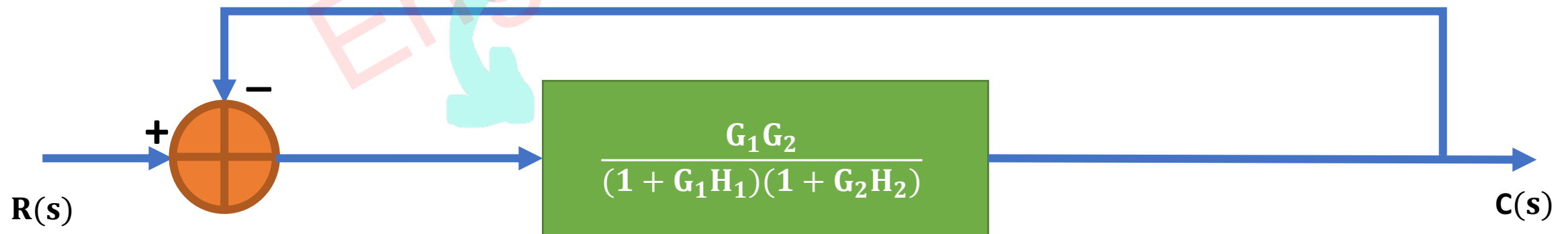
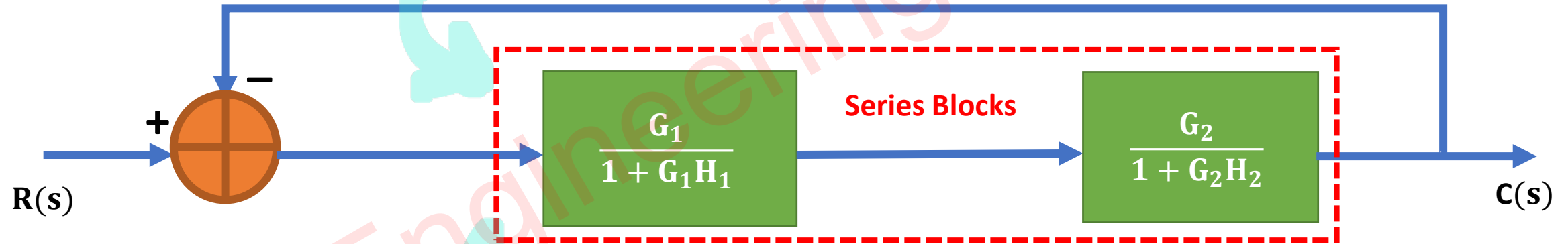
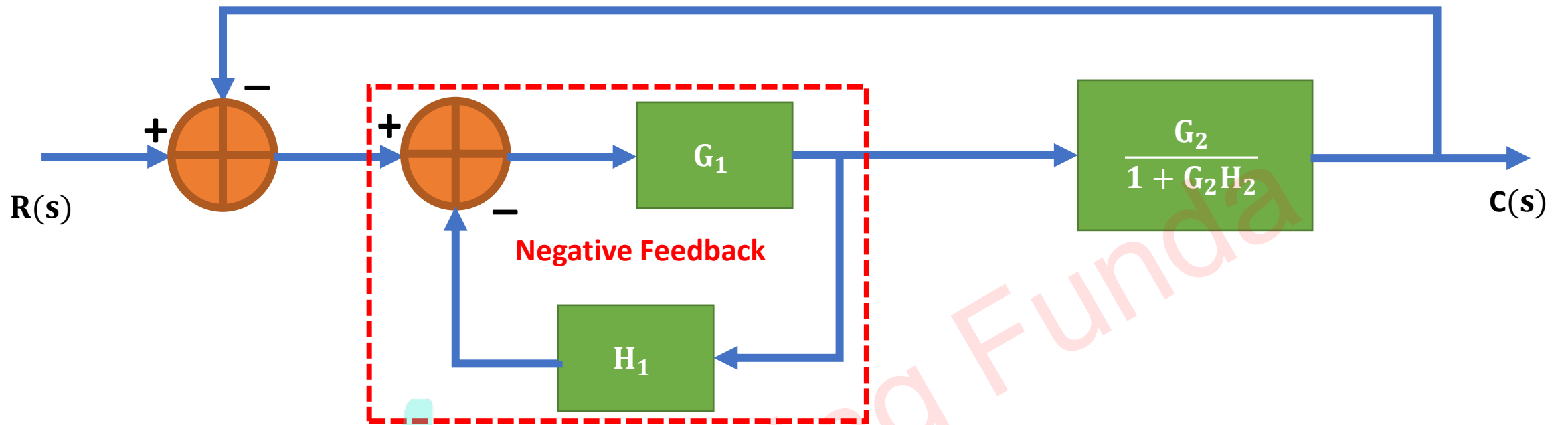
Order of Block Diagram Reduction Rules

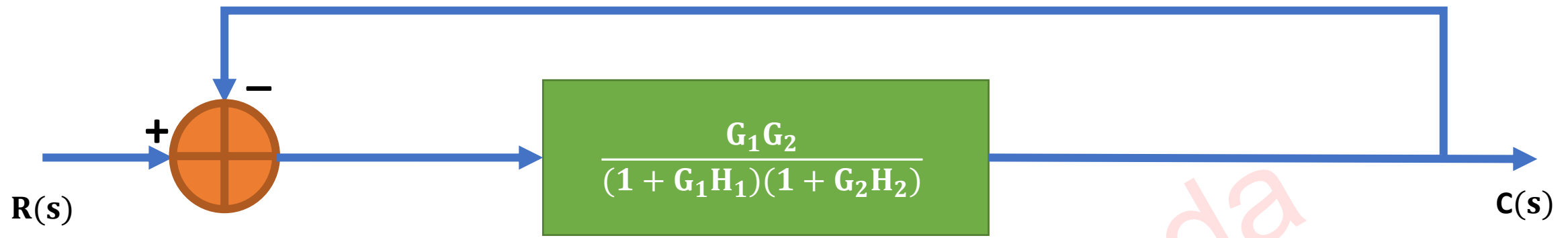
1. Priority 1: Check for series and parallel connections and place the equivalent.
2. Priority 2: Check for negative or positive feedback.
3. Priority 3: Forward/Backward **Summing point** to make above priority 1 and 2.
4. Priority 3: Forward/Backward **Branch point** to make above priority 1 and 2.
5. Priority 3: Interchange Summing Point to make above priority 1 and 2.
6. Priority 3: Split Summing Points to make above priority 1 and 2.

Question 1 – Find the Transfer Function of a given system.









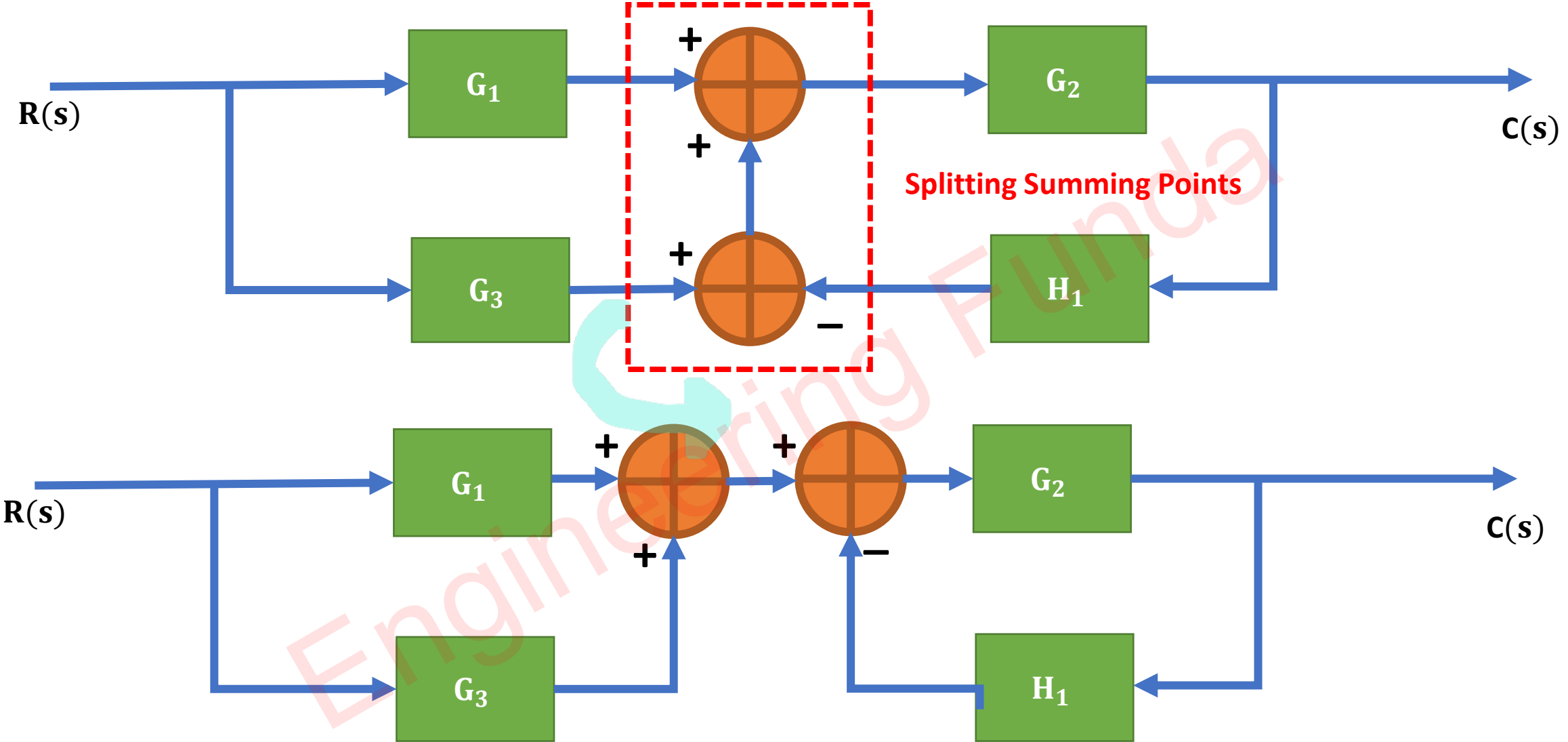
Negative Feedback

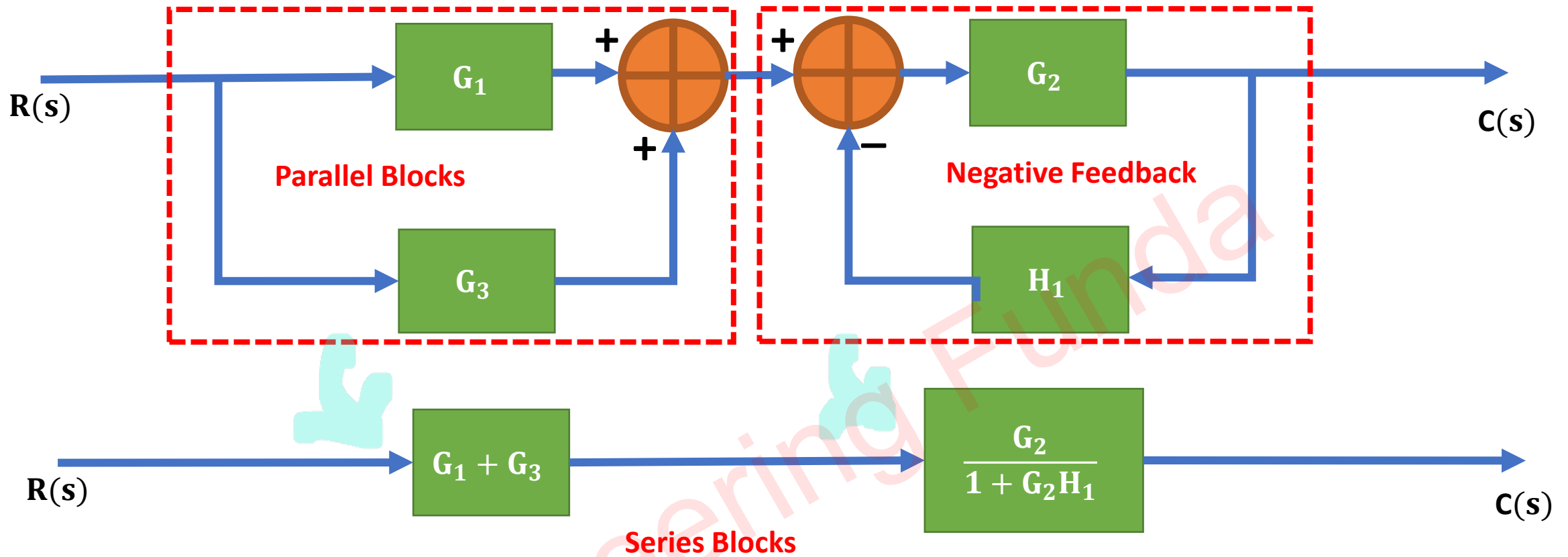
$$\Rightarrow \frac{C(S)}{R(S)} = \frac{G(s)}{1 + G(s)H(s)}$$

$$\Rightarrow H(s) = 1, G(s) = \frac{G_1 G_2}{(1 + G_1 H_1)(1 + G_2 H_2)}$$

$$\Rightarrow \frac{C(S)}{R(S)} = \frac{G_1 G_2}{1 + G_1 G_2 + G_1 H_1 + G_2 H_2 + G_1 G_2 H_1 H_2}$$

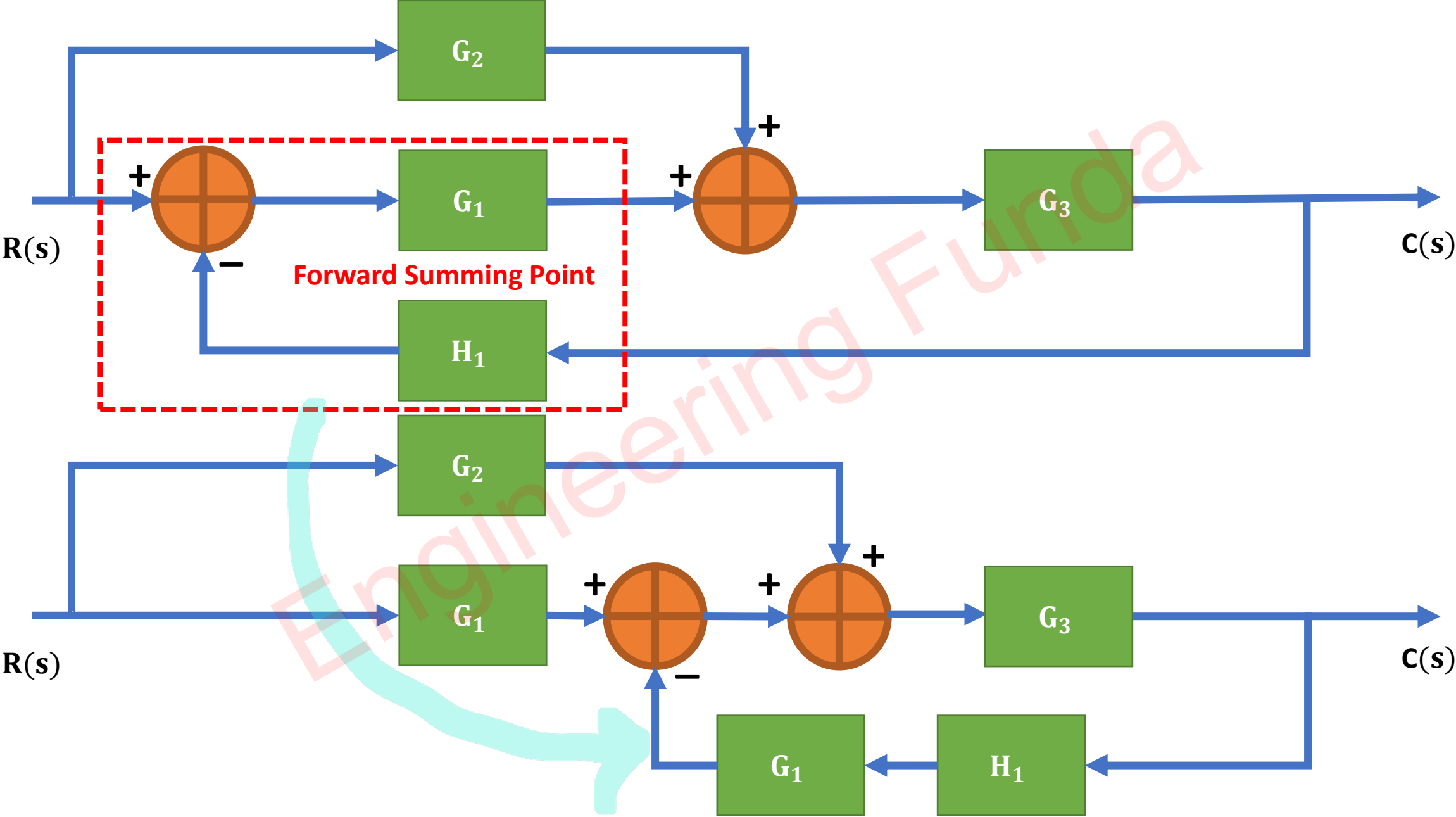
Question 2 – Find the Transfer Function of a given system.

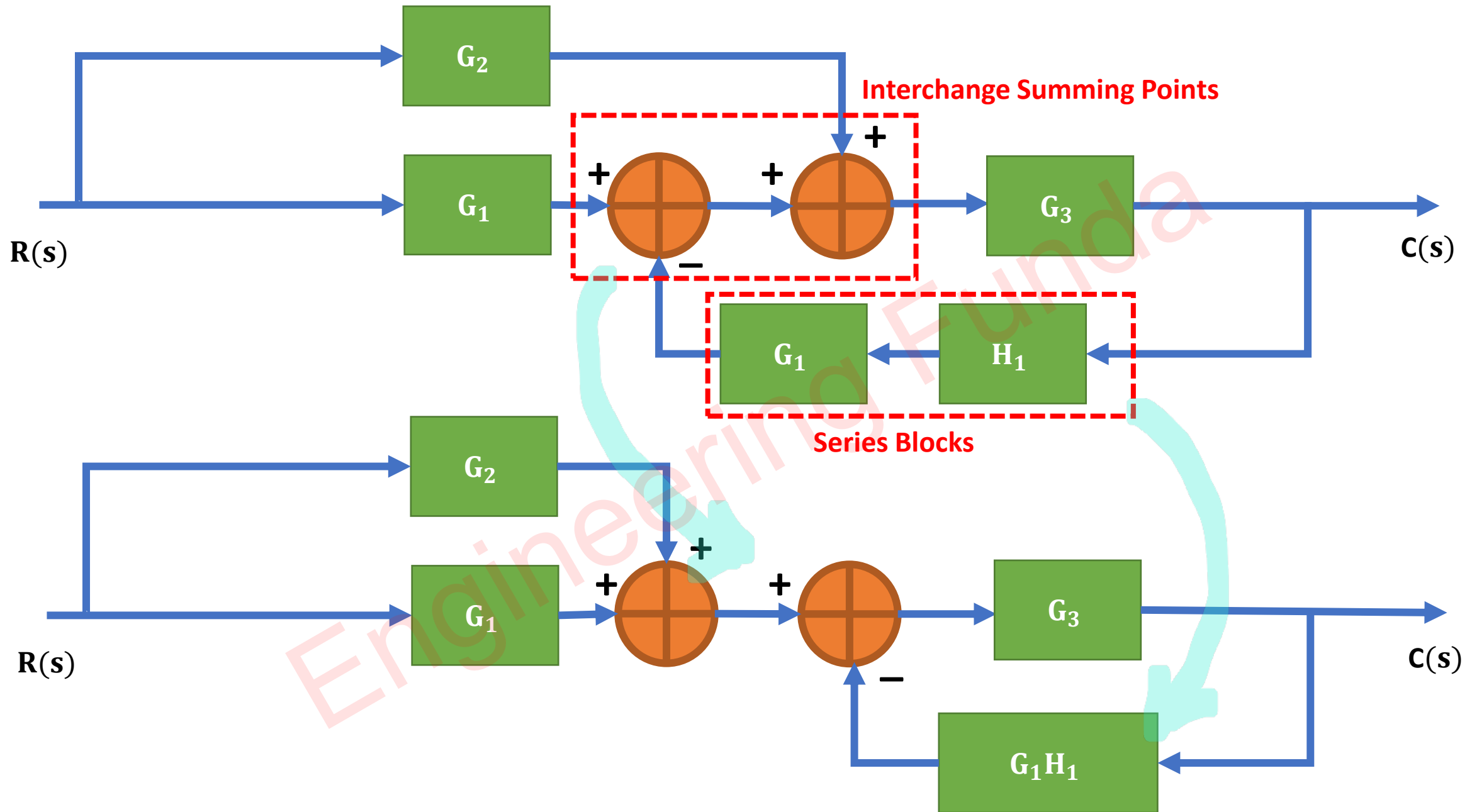


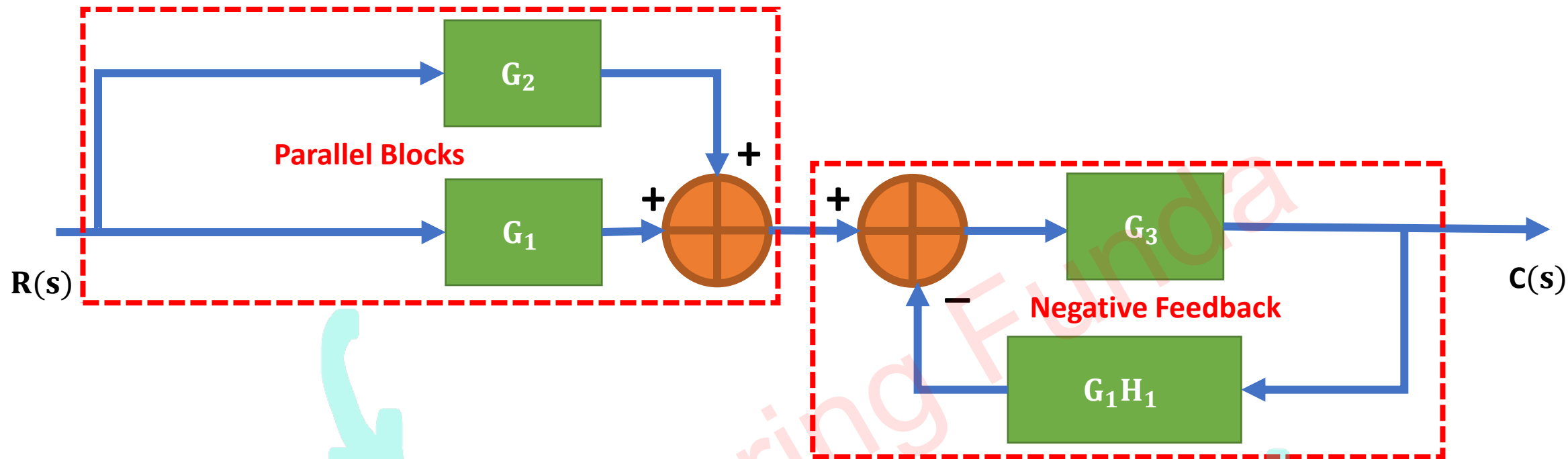


$$\Rightarrow \frac{C(S)}{R(S)} = (G_1 + G_3) \left(\frac{G_2}{1 + G_2 H_1} \right)$$

Question 3 – Find the Transfer Function of a given system.







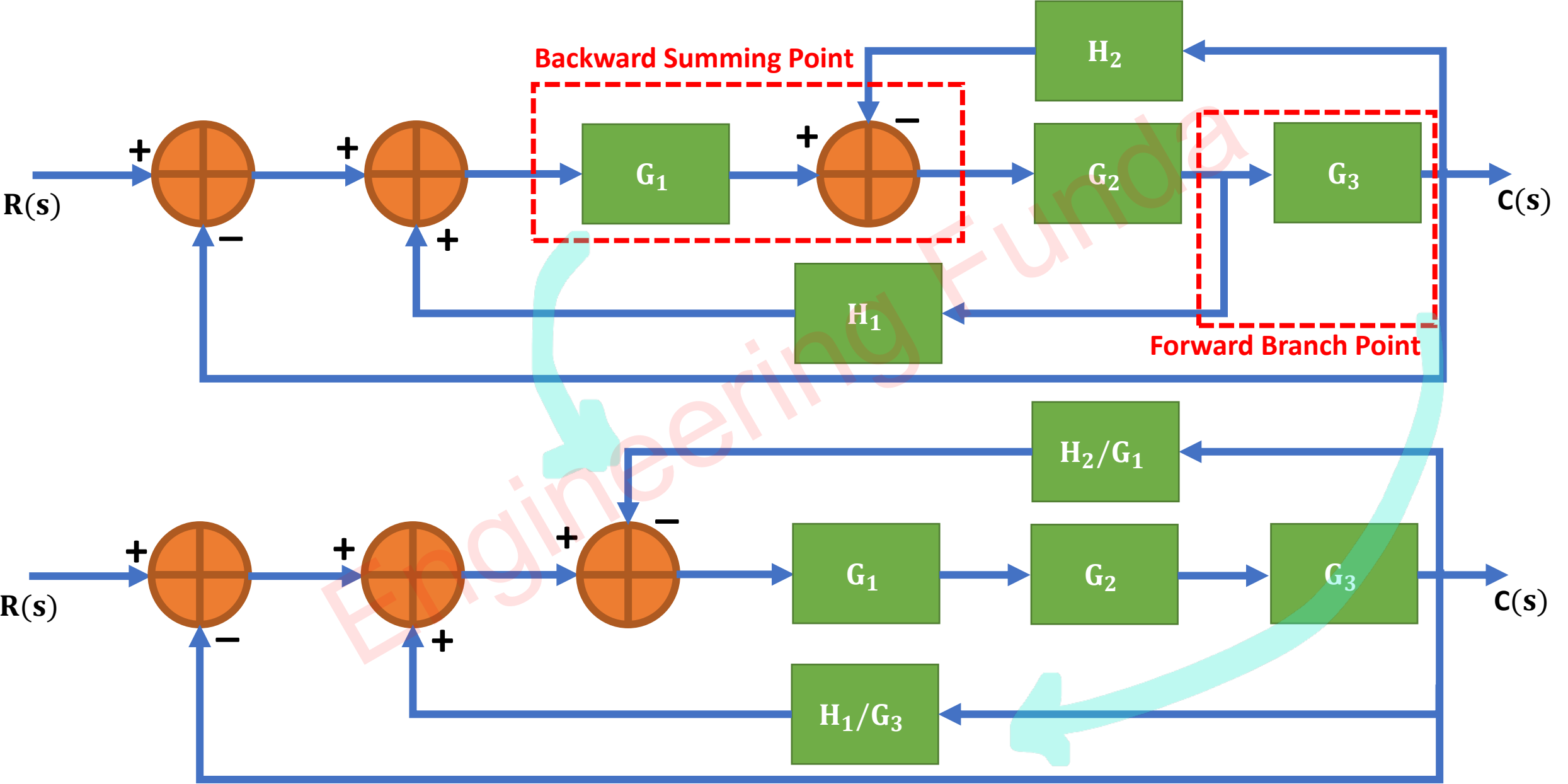
$$\Rightarrow \frac{C(S)}{R(S)} = (G_1 + G_2) \left(\frac{G_3}{1 + G_1G_3H_1} \right)$$

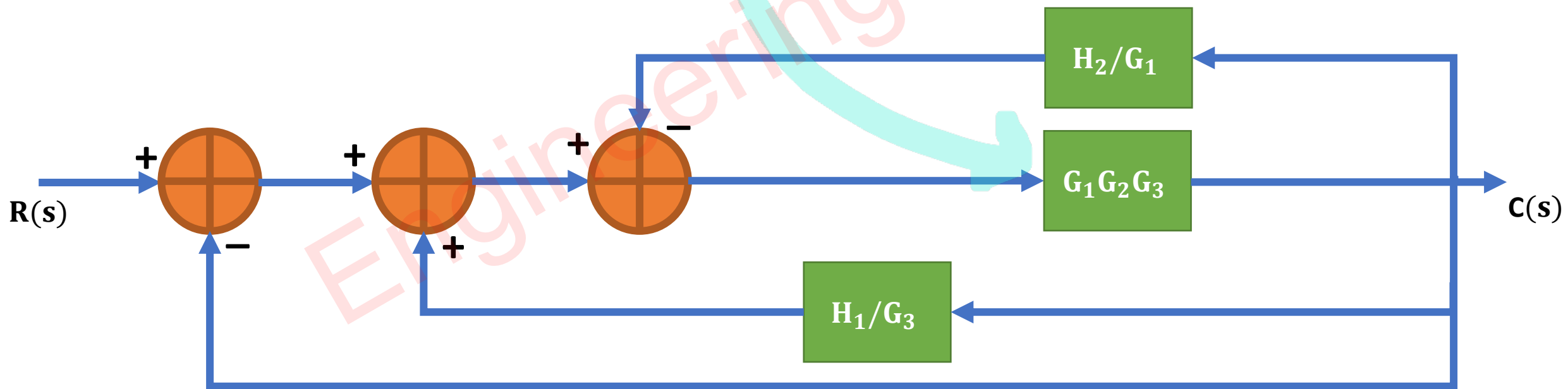
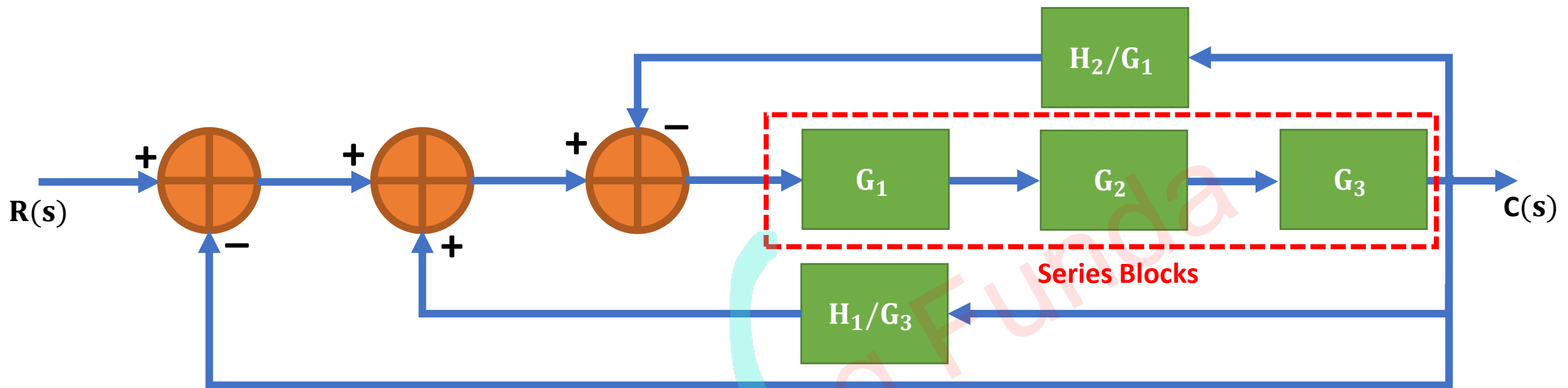
Block Diagram Reduction Examples

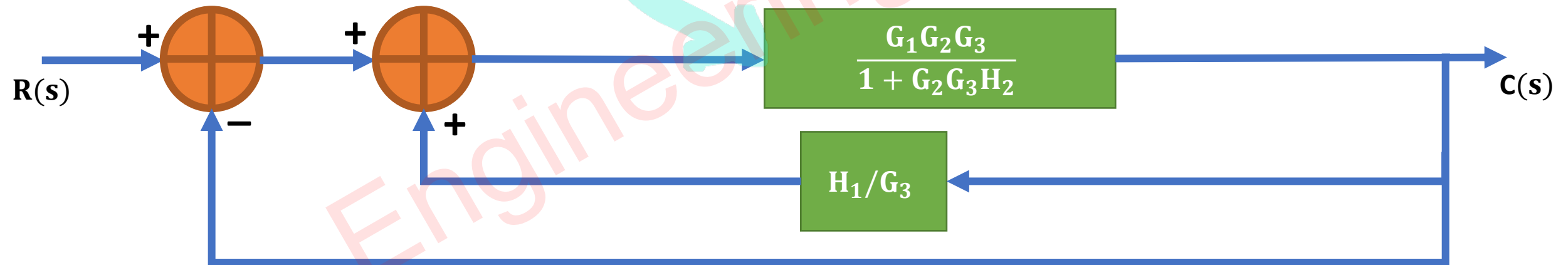
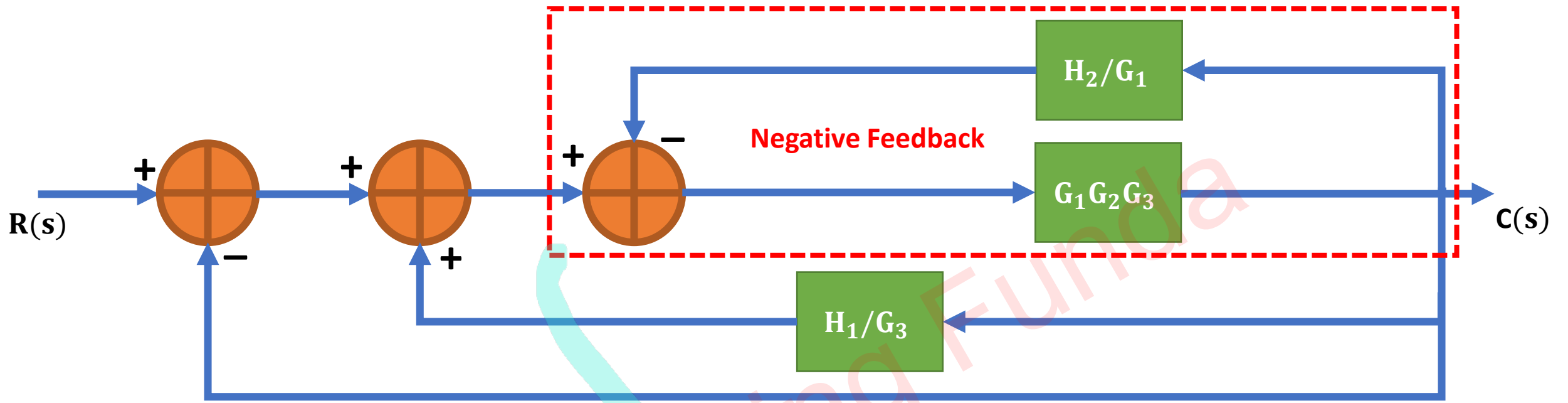
Order of Block Diagram Reduction Rules

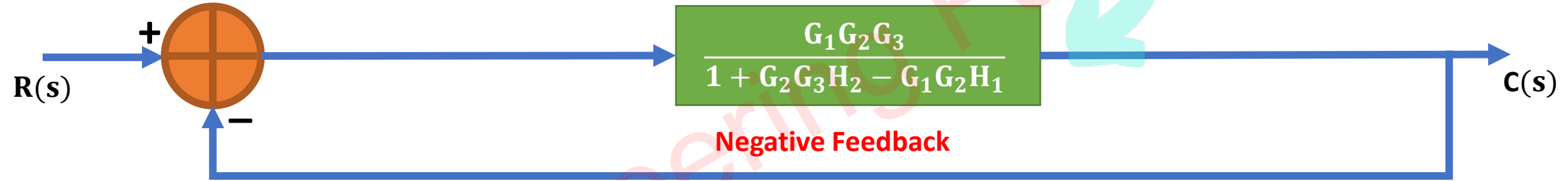
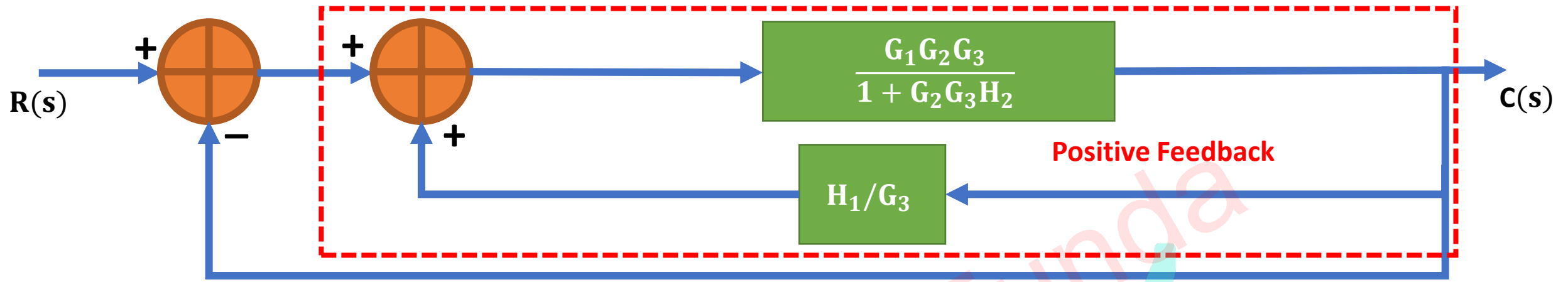
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6. Priority 3: Split Summing Points to make above priority 1 and 2.

Question 4 – Find the Transfer Function of a given system.



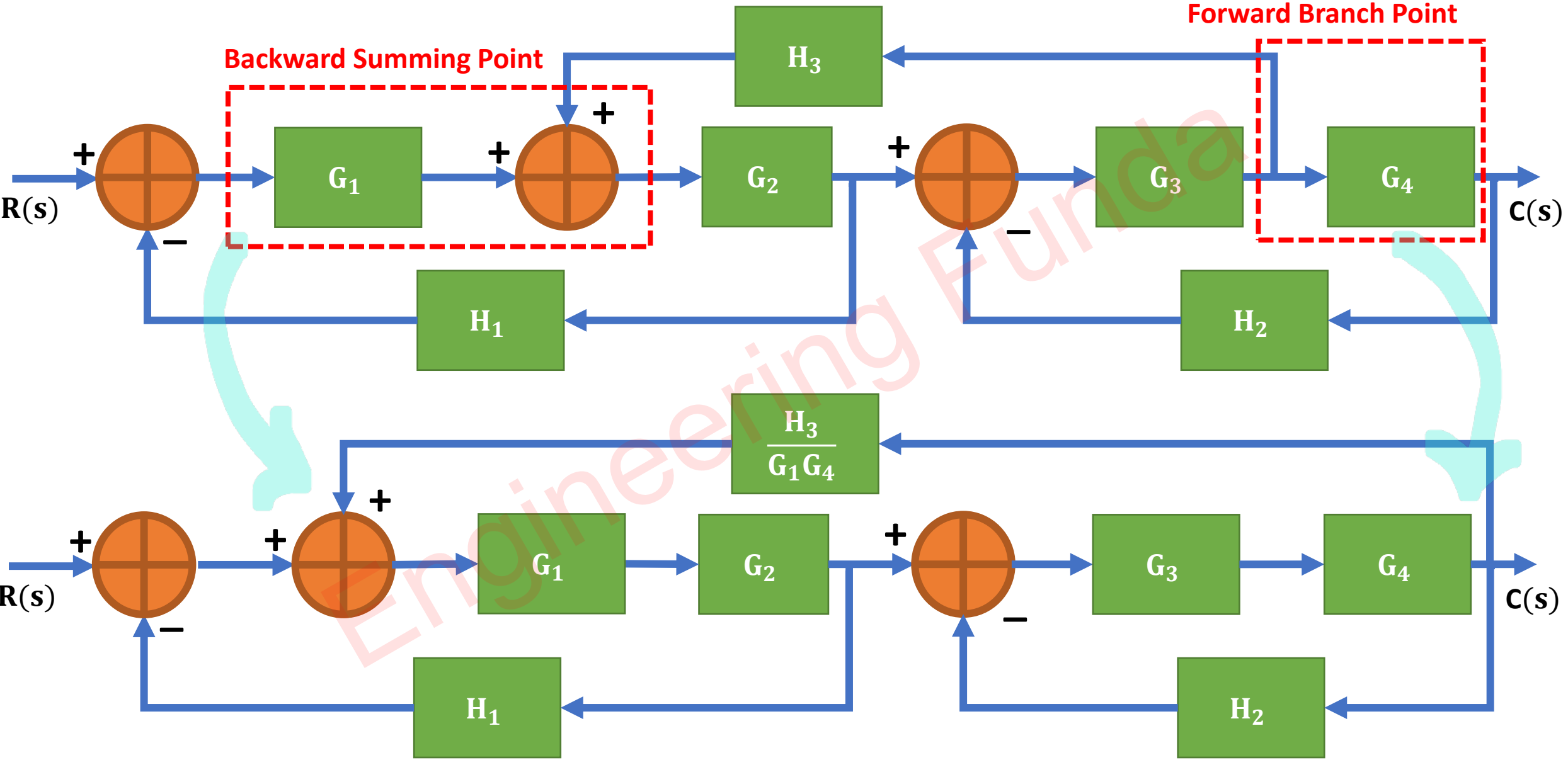


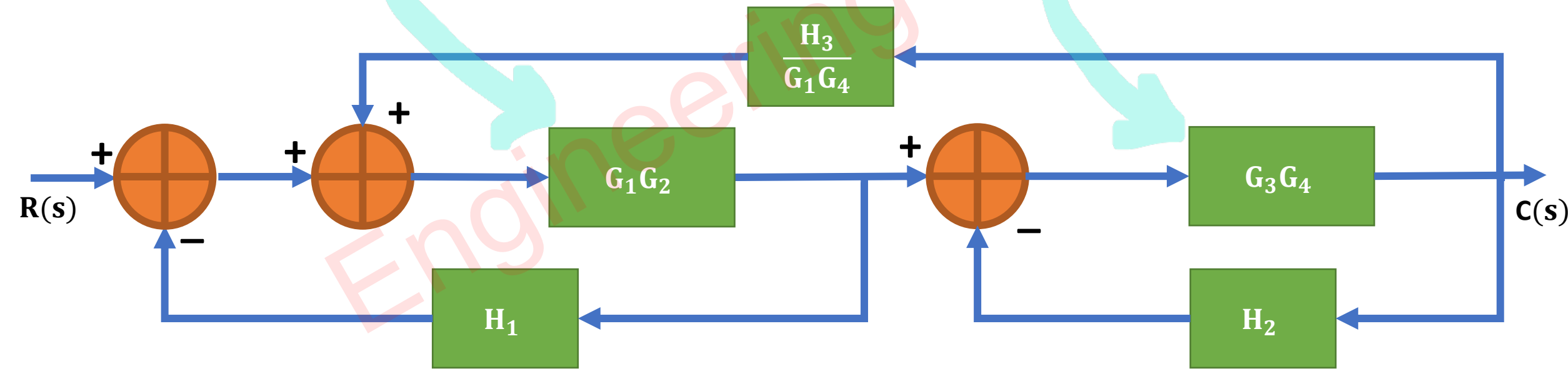
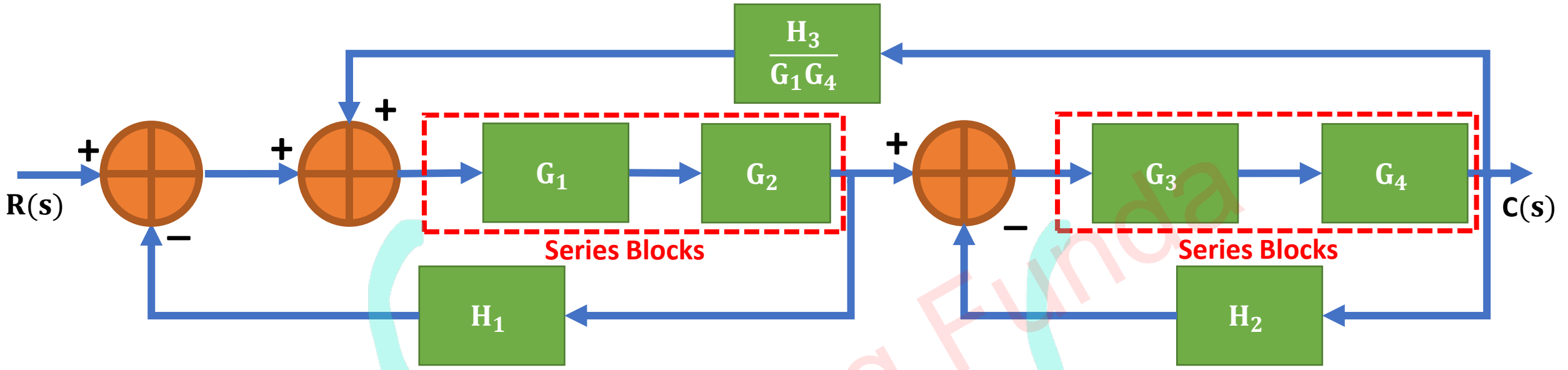




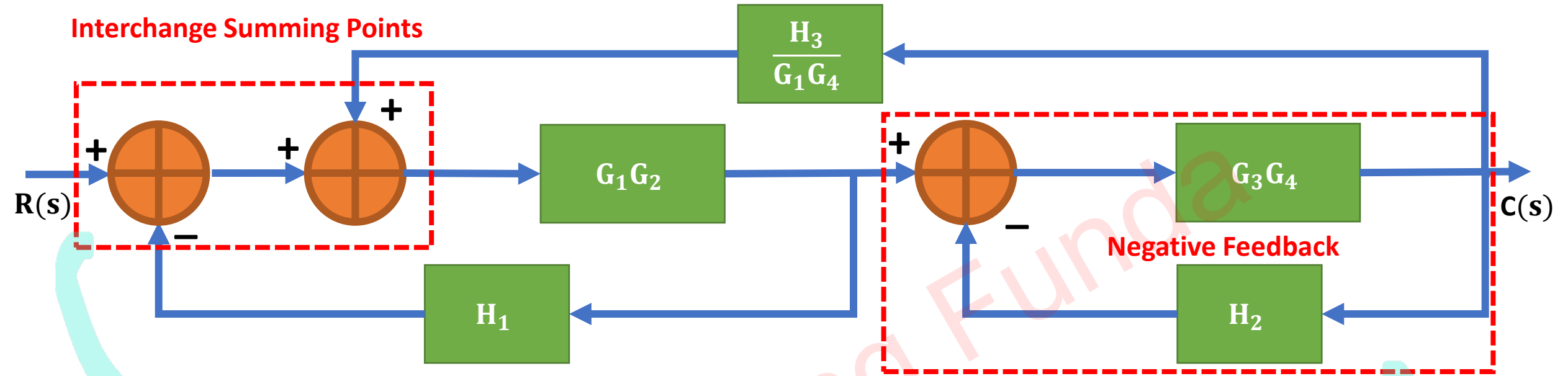
$$\Rightarrow \frac{C(S)}{R(S)} = \frac{G_1 G_2 G_3}{1 + G_2 G_3 H_2 - G_1 G_2 H_1 + G_1 G_2 G_3}$$

Question 5 – Find the Transfer Function of a given system.

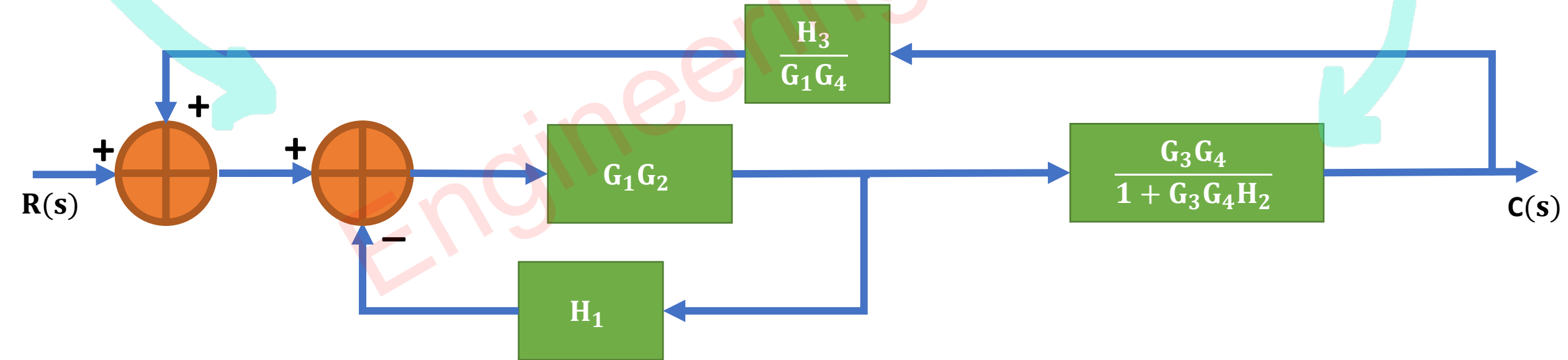


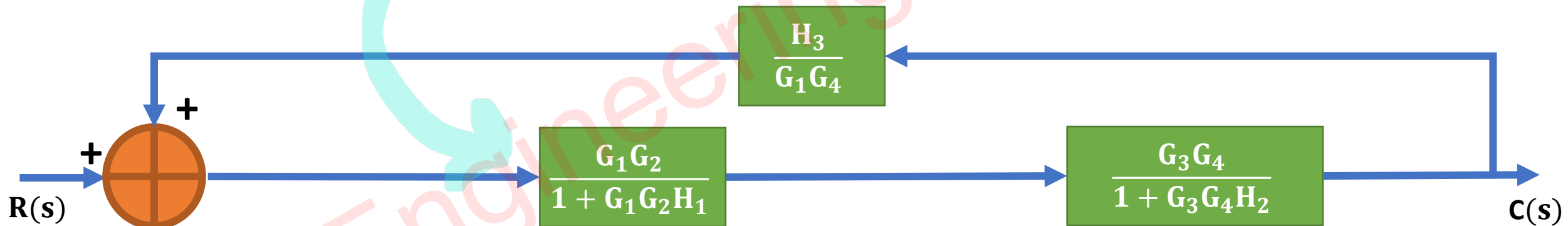
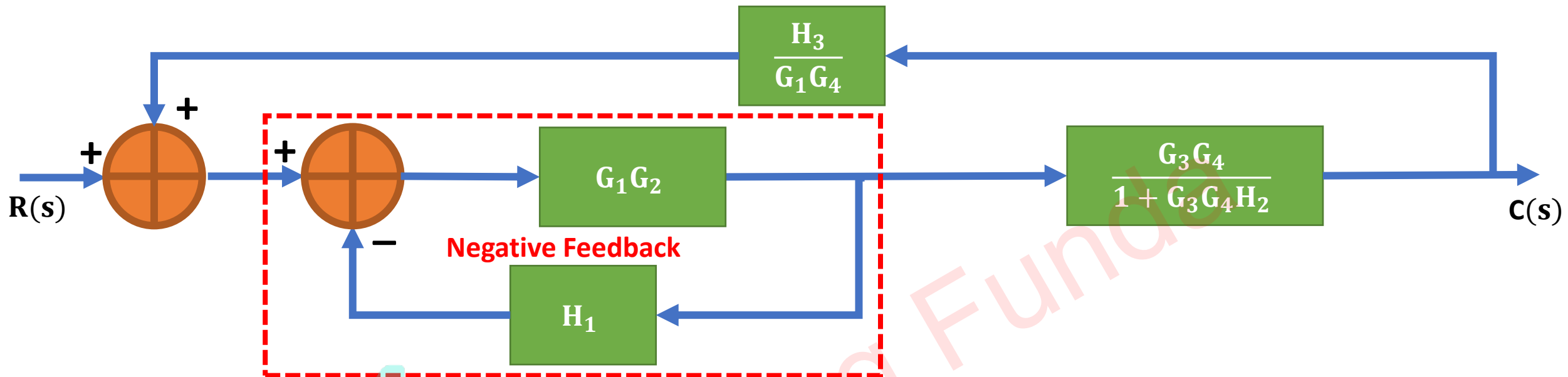


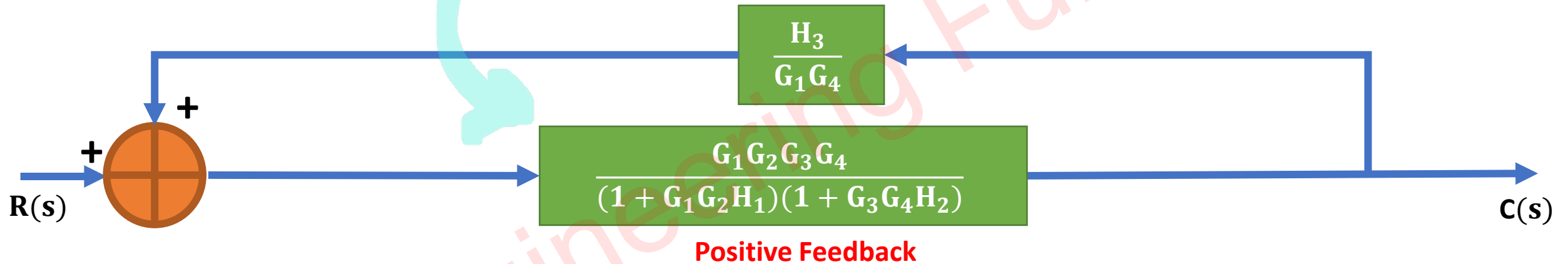
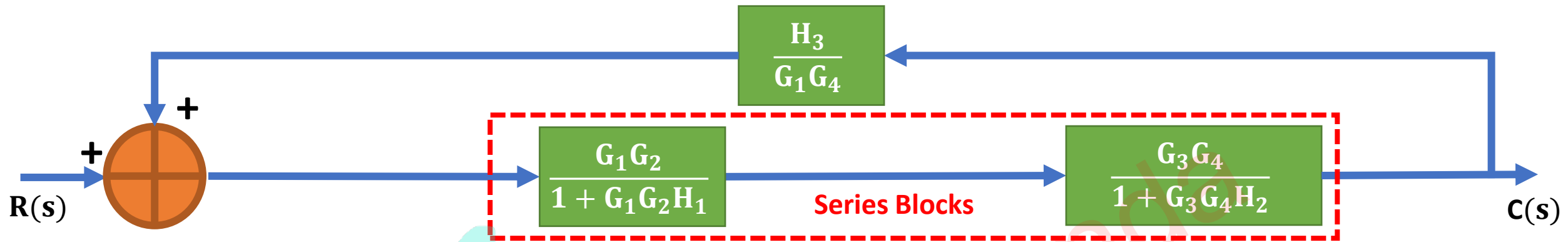
Interchange Summing Points



Negative Feedback

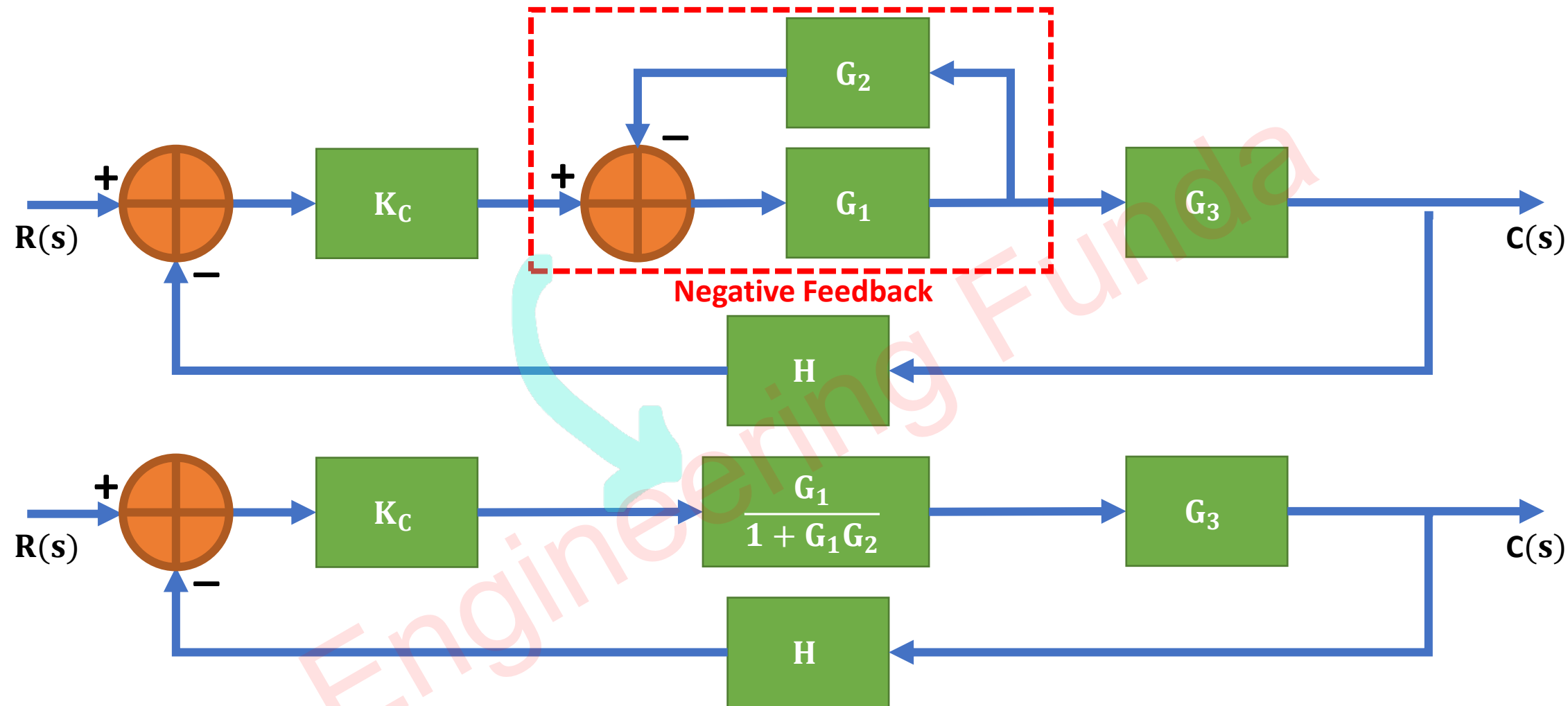


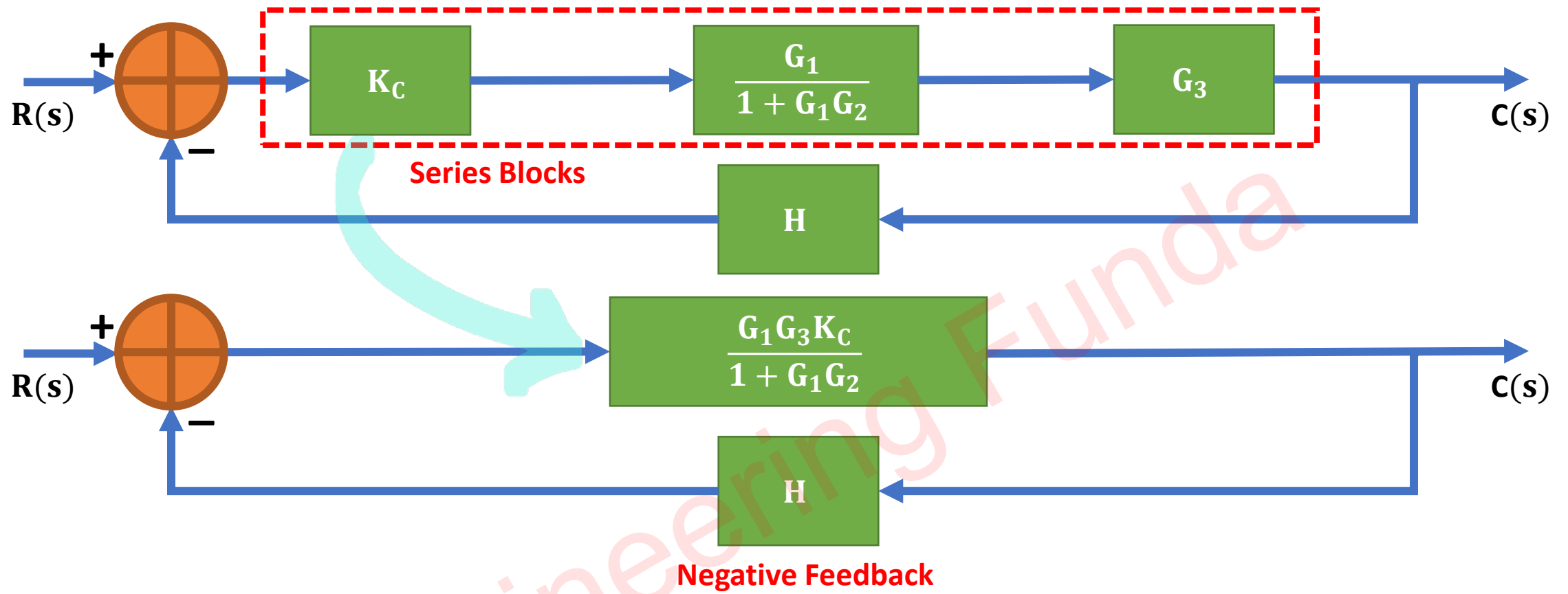




$$\Rightarrow \frac{C(S)}{R(S)} = \frac{G_1 G_2 G_3 G_4}{1 + G_1 G_2 H_1 + G_3 G_4 H_2 - G_2 G_3 H_3 + G_1 G_2 G_3 G_4 H_1 H_2}$$

Question 6 – Find the Transfer Function of a given system.





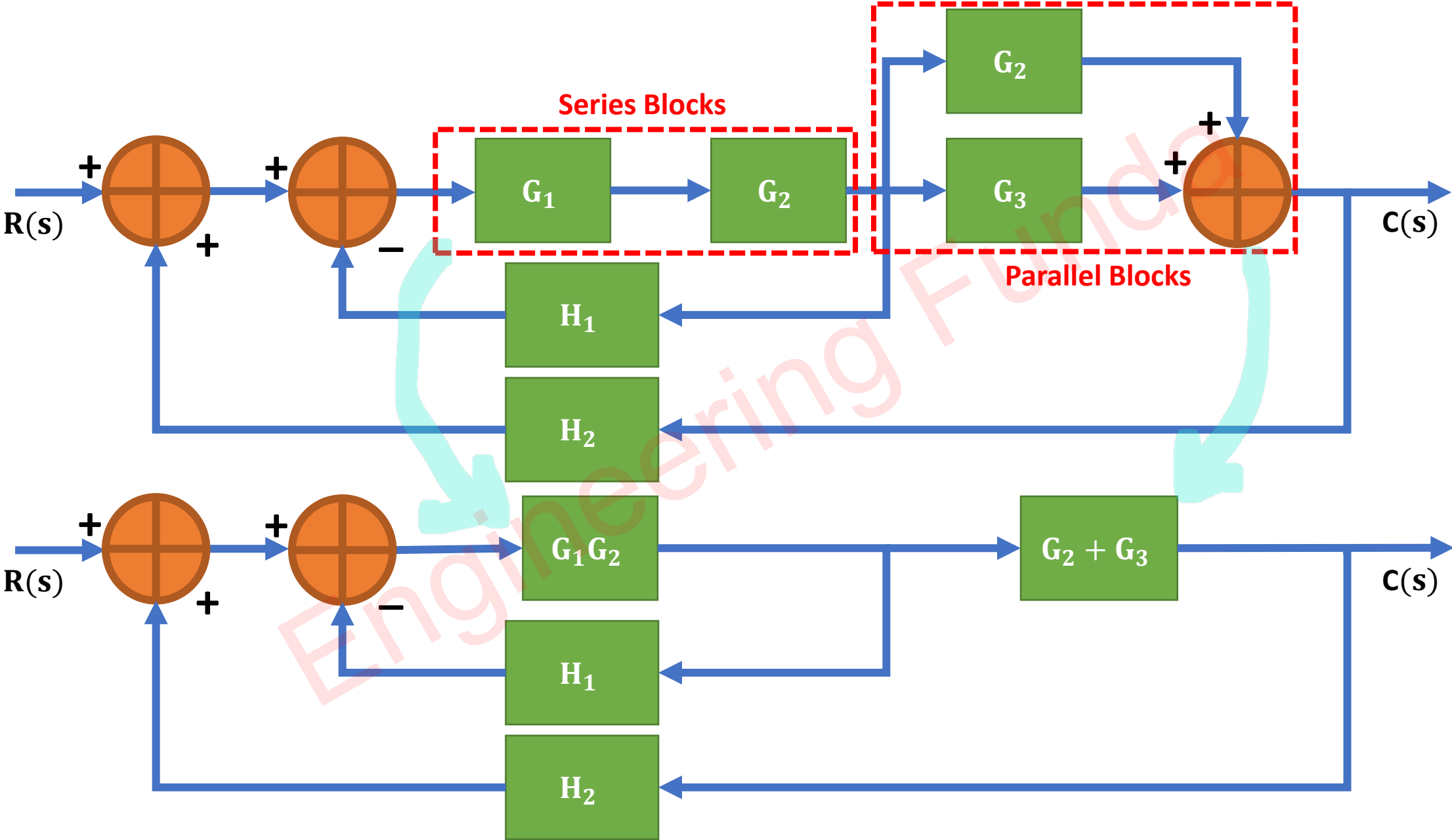
$$\Rightarrow \frac{C(S)}{R(S)} = \frac{G_1 G_3 K_C}{1 + G_1 G_2 + G_1 G_3 K_C H}$$

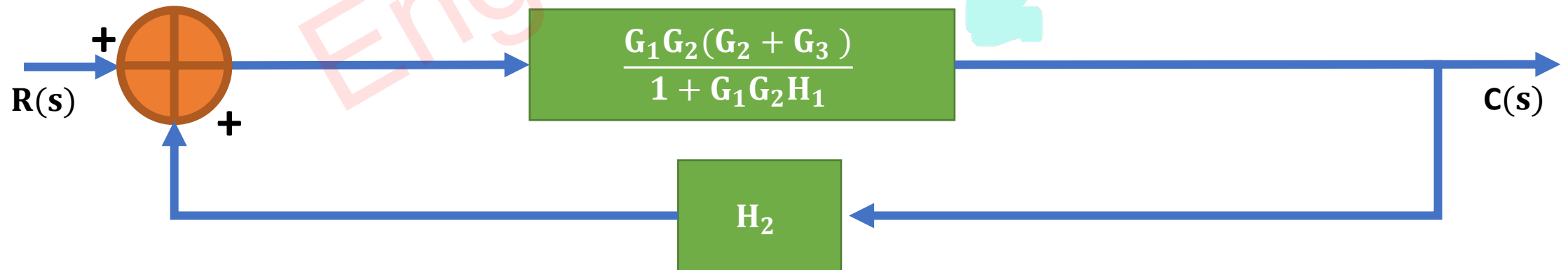
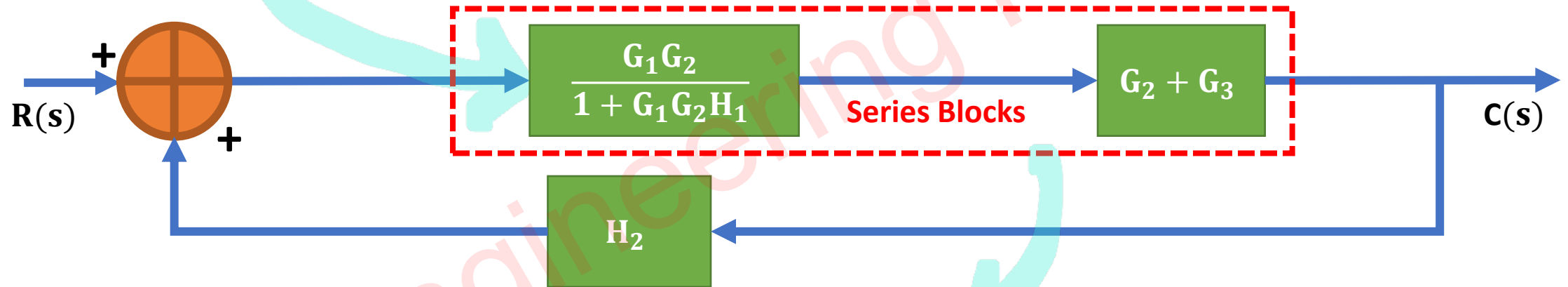
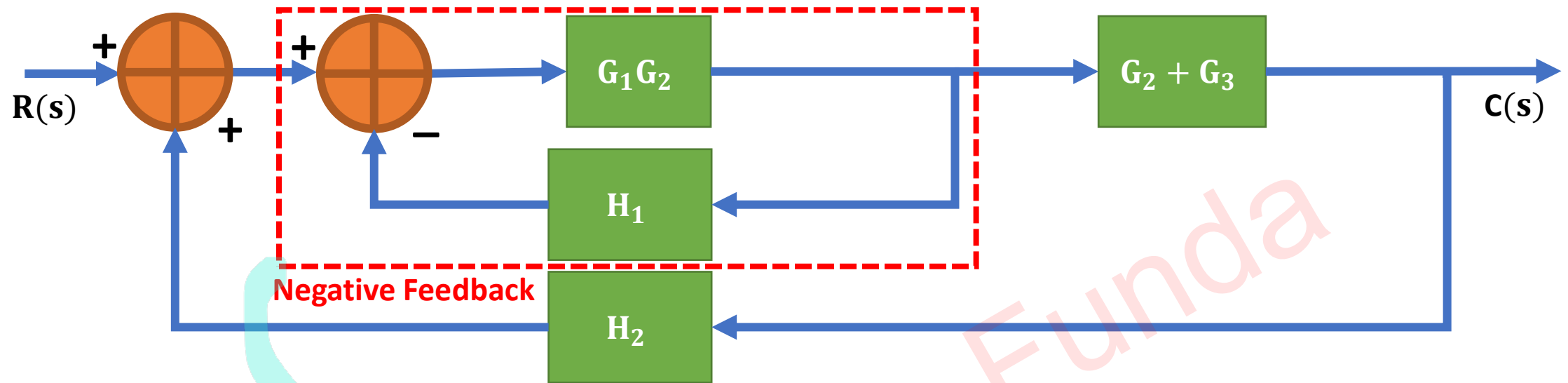
Block Diagram Reduction Examples

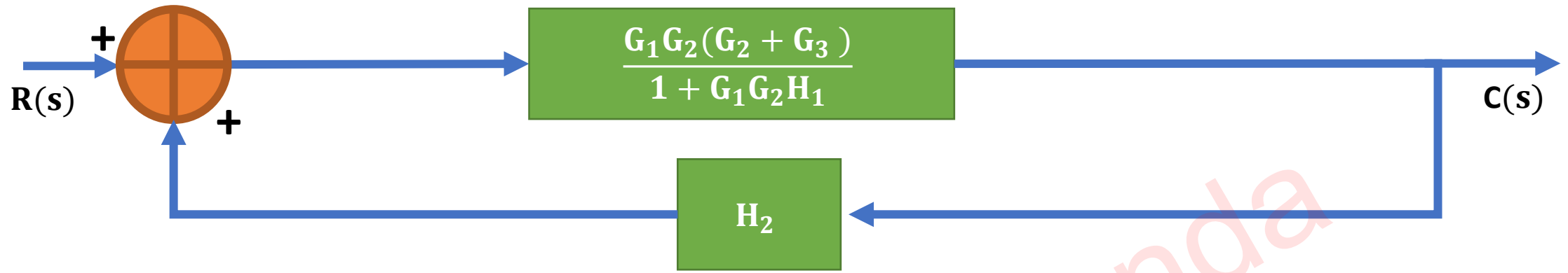
Order of Block Diagram Reduction Rules

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6. Priority 3: Split Summing Points to make above priority 1 and 2.

Question 7 – Find the Transfer Function of a given system.



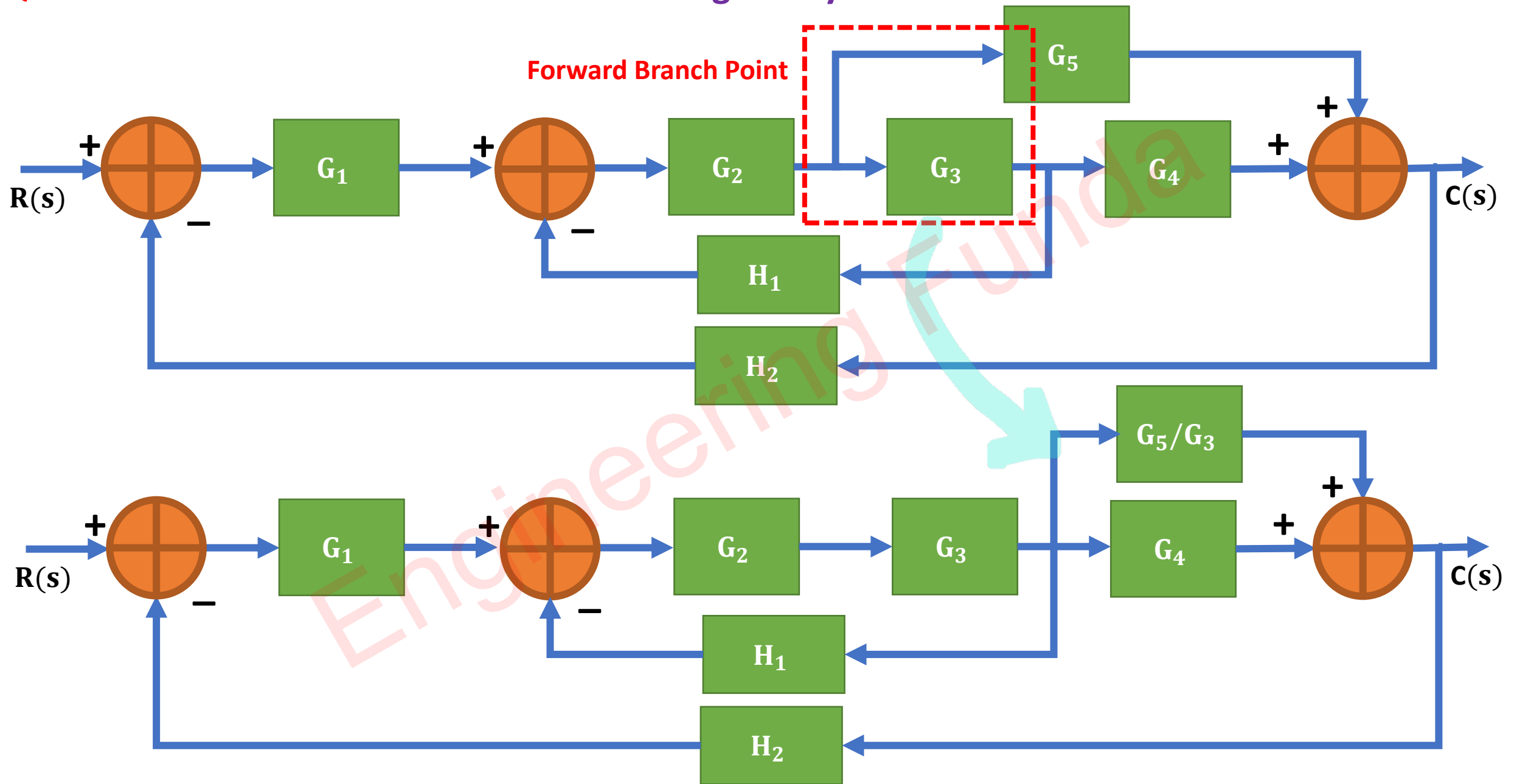


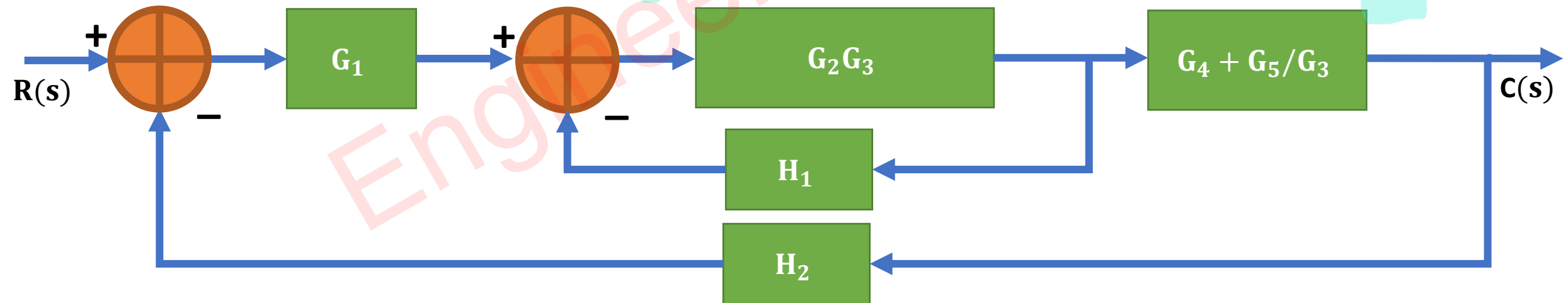
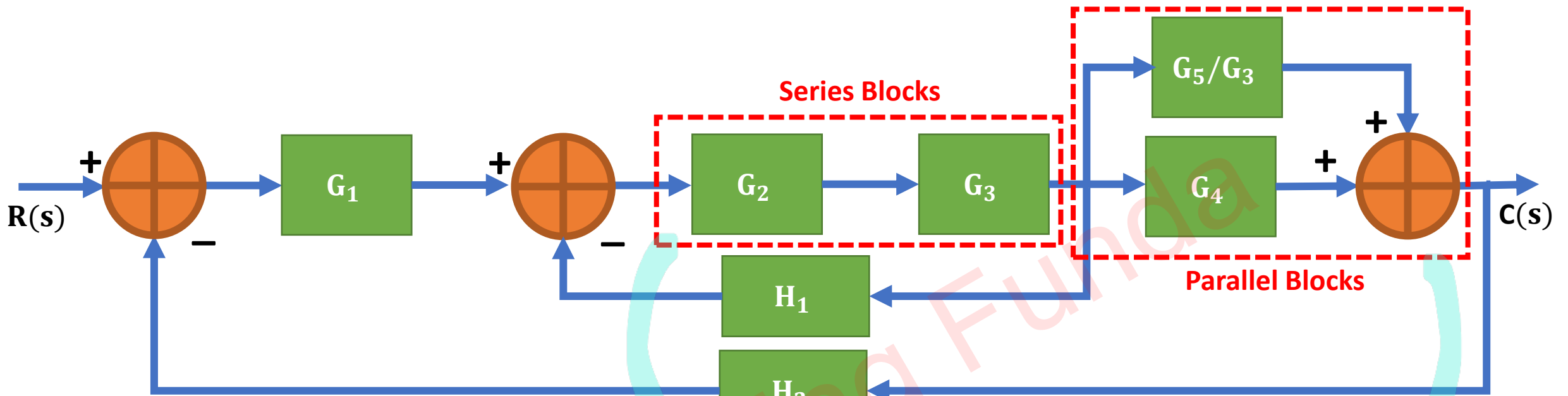


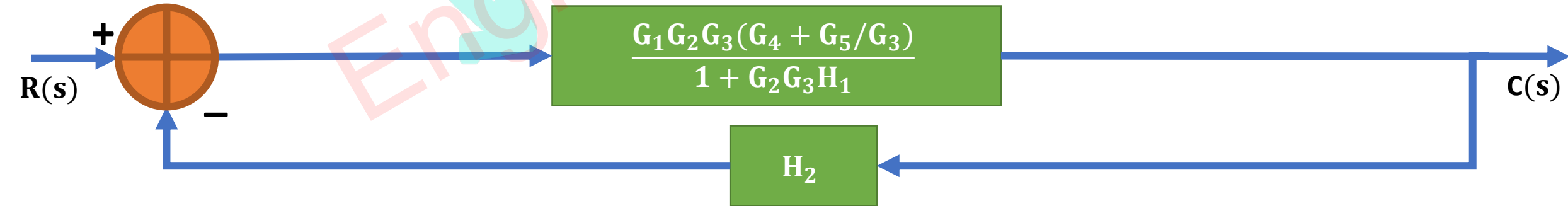
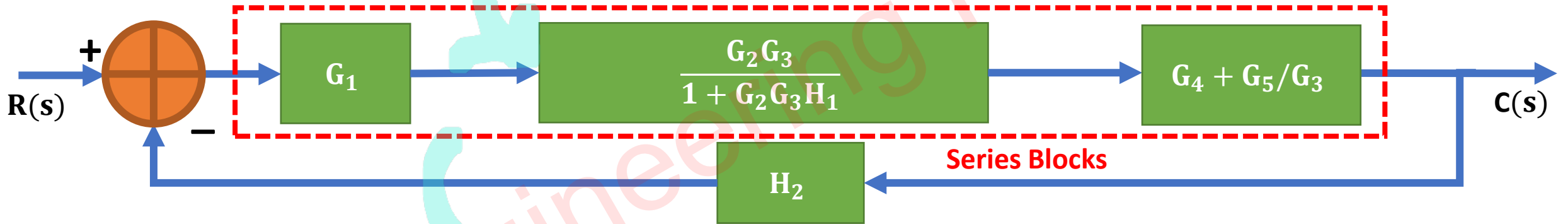
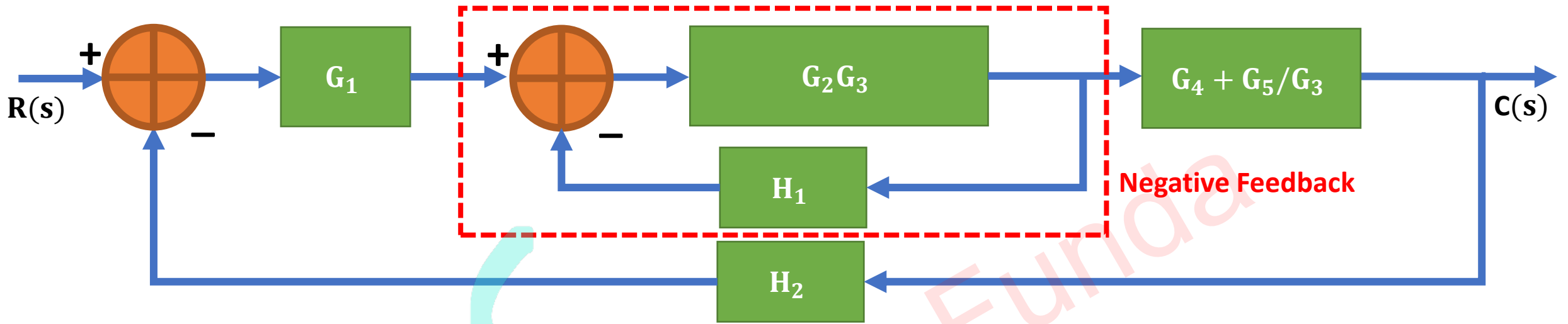
Positive Feedback

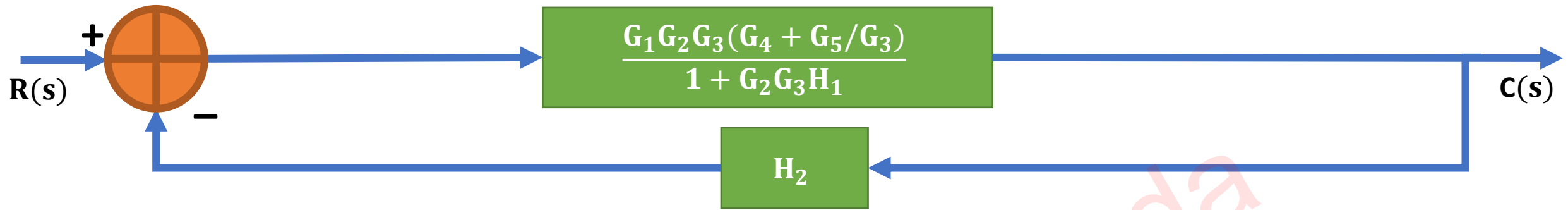
$$\Rightarrow \frac{C(S)}{R(S)} = \frac{G_1 G_2 (G_2 + G_3)}{1 + G_1 G_2 H_1 - H_2 G_1 G_2 (G_2 + G_3)}$$

Question 8 – Find the Transfer Function of a given system.







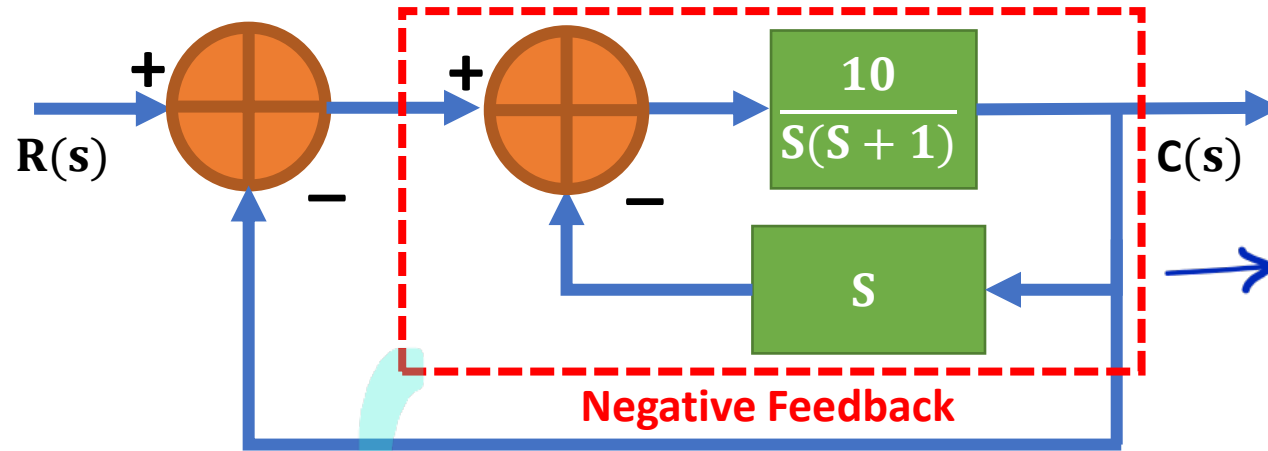


Negative Feedback

$$\Rightarrow \frac{C(S)}{R(S)} = \frac{K}{1 + KH_2}$$

$$\Rightarrow \text{Where, } K = \frac{G_1 G_2 G_3 (G_4 + G_5/G_3)}{1 + G_2 G_3 H_1}$$

Question 9 – Find the Transfer Function of a given system.



$$\Rightarrow \frac{C(S)}{R(S)} = \frac{10}{S^2 + 11S + 10}$$

$$\begin{aligned} &\rightarrow = \frac{10/S(S+1)}{1 + S(10/S(S+1))} \\ &= \frac{10}{S(S+1) + 10S} = \frac{10}{S^2 + 11S} \end{aligned}$$

$$\rightarrow = \frac{10/(S^2 + 11S)}{1 + 10/(S^2 + 11S)} = \frac{10}{S^2 + 11S + 10}$$